

RPSC · RAS & RTS PRELIMINARY EXAMINATION 2026

Geography of the World

185 practice questions across 5 chapters, each chapter opening with a must-know fact sheet. Answers and explanations are at the back, so you can work through the paper first.

WG01	Physical Geography: Mountains, Plateaus, Plains, Deserts	40
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185 questions · **141** fact-sheet points · **22** tagged as previously asked
Paper pattern: 150 questions · 200 marks · 3 hours · 1/3 negative marking

WG01

Physical Geography: Mountains, Plateaus, Plains, Deserts

40 questions · 30 fact-sheet points

Must-know facts

1. Mount Everest (8,848.86 m), Great Himalaya, Nepal-China border - highest peak on earth and of Asia.
2. Aconcagua (about 6,960 m), Andes, Argentina - highest peak of South America and of the whole Western and Southern Hemisphere.
3. Denali (6,190 m), Alaska Range, USA - highest peak of North America; formerly called Mount McKinley.
4. Kilimanjaro (5,895 m), Tanzania - highest peak of Africa; a dormant stratovolcano. Mount Kenya (5,199 m) is second.
5. Mount Elbrus (5,642 m), Caucasus, Russia - highest peak of Europe. Mont Blanc (4,808 m), Alps, is highest in Western Europe.
6. Vinson Massif (4,892 m), Sentinel Range - highest peak of Antarctica.
7. Puncak Jaya / Carstensz Pyramid (4,884 m), New Guinea, Indonesia - highest of Oceania; Mount Kosciuszko (2,228 m) is highest on mainland Australia.
8. Descending order of continental highest peaks: Everest > Aconcagua > Denali > Kilimanjaro > Elbrus > Vinson Massif > Puncak Jaya.
9. K2 / Godwin Austen (8,611 m), Karakoram - world's second highest; Kangchenjunga (8,586 m) third.
10. Andes (about 7,000 km) - longest mountain range on land; the Mid-Oceanic Ridge (about 65,000 km) is the longest range overall.
11. Ural Mountains - conventional boundary between Europe and Asia. Atlas - north-west Africa; Drakensberg - South Africa; Great Dividing Range - eastern Australia.
12. Tibetan Plateau - highest and largest plateau, 'Roof of the World', average elevation above 4,500 m, between the Himalaya and the Kunlun.
13. Plateau types: intermontane (Tibet, Bolivia, Anatolia), lava/volcanic (Deccan, Columbia), dissected (Colorado - Grand Canyon), continental (Chotanagpur), piedmont, rain-shadow (Patagonia).
14. Plateau-country pairs: Katanga - DR Congo (copper); Anatolian - Turkey; Meseta - Spain; Mato Grosso - Brazil; Loess Plateau - China.
15. Earth's interior: crust-mantle boundary = Mohorovicic discontinuity; mantle-core = Gutenberg discontinuity. Layers by composition: SIAL, SIMA, NIFE.
16. S-waves travel only through solids; their shadow zone beyond 103 degrees proves the outer core is liquid. P-waves pass through solids and liquids.
17. Continental drift - Alfred Wegener (1912): Pangaea surrounded by Panthalassa, splitting into Laurasia and Gondwana across the Tethys Sea. Sea-floor spreading - Harry Hess.
18. Circum-Pacific belt or 'Ring of Fire' - about two-thirds of the world's active volcanoes and the great majority of earthquakes; dominated by subduction zones.
19. Ojos del Salado (Chile-Argentina) - world's highest volcano; Mauna Loa (Hawaii) - largest active volcano; Etna - Europe's most active; Stromboli - 'Lighthouse of the Mediterranean'.
20. Largest deserts: Antarctic (about 13.9 m sq km) and Arctic (about 13.7 m sq km) polar deserts; Sahara (over 8.5 m sq km) is the largest hot desert.
21. Desert-continent pairs: Gobi - Asia; Kalahari and Namib - southern Africa; Atacama and Patagonian - South America; Great Victoria and Great Sandy - Australia; Great Basin, Sonoran, Chihuahuan, Mojave - North America.

22. Atacama (Chile) - driest desert, kept arid by the cold Peru current and the Andean rain shadow; Namib - oldest desert; Patagonian - largest desert of the Americas; Great Victoria - largest in Australia.
23. Asian deserts: Arabian (with the Rub al-Khali), Karakum (Turkmenistan), Kyzylkum (Uzbekistan-Kazakhstan), Taklamakan (China), Thar (India-Pakistan), Kavir (Iran).
24. Wind erosion landforms: yardang, zeugen, inselberg, ventifact, mushroom rock, pediment. Wind deposition: barchan (crescent, horns downwind), seif/longitudinal dune, loess.
25. Glacial erosion: cirque, arete, horn, U-shaped valley, hanging valley, fjord (drowned glacial valley). Glacial deposition: moraine, drumlin, esker, outwash plain.
26. Atmosphere upward: troposphere (all weather; 18 km at equator, 8 km at poles) - stratosphere (ozone) - mesosphere (coldest, meteors burn) - thermosphere/ionosphere - exosphere. Normal lapse rate 6.5 C per km.
27. Pressure belts: equatorial low (doldrums), sub-tropical high (horse latitudes), sub-polar low, polar high. Planetary winds: Trades, Westerlies (Roaring Forties), Polar Easterlies.
28. Local winds: Chinook (Rockies, 'snow-eater'), Foehn (Alps), Mistral and Bora (cold), Sirocco (Sahara to Mediterranean), Harmattan (West Africa, 'the Doctor'), Santa Ana (California), Zonda (Argentina), Loo (north India).
29. Warm currents: Gulf Stream, North Atlantic Drift, Kuroshio, Brazil, Agulhas, East Australian. Cold currents: Labrador, Canary, Benguela, Peru (Humboldt), California, Oyashio, West Wind Drift.
30. Spring tides on new and full moon (syzygy); neap tides at quadrature. Bay of Fundy (Canada) has the world's highest tidal range. Great Barrier Reef (Coral Sea, Australia) is the largest coral reef system; reef types - fringing, barrier, atoll.

1. Which of the following is the highest peak of the African continent? RAS PRE 2018
 - (A) Mount Kenya
 - (B) Mount Kilimanjaro
 - (C) Ras Dashen
 - (D) Mount Stanley
2. Aconcagua, the highest peak of South America, lies in which mountain range? RAS PRE 2016
 - (A) Andes
 - (B) Sierra Madre
 - (C) Rocky Mountains
 - (D) Atlas Mountains
3. Mount Elbrus, the highest peak of Europe, is located in which mountain range?
 - (A) Pyrenees
 - (B) Carpathians
 - (C) Alps
 - (D) Caucasus
4. Vinson Massif is the highest peak of which continent? RAS PRE 2021
 - (A) Antarctica
 - (B) Australia
 - (C) South America
 - (D) Europe
5. Denali, the highest mountain of North America, is situated in which country?
 - (A) Mexico
 - (B) United States of America
 - (C) Greenland
 - (D) Canada
6. Which is the longest mountain range on the surface of the continents?
 - (A) Great Dividing Range
 - (B) Himalayas
 - (C) Rocky Mountains
 - (D) Andes
7. Which mountain range is conventionally taken as the boundary between Europe and Asia?
 - (A) Carpathians
 - (B) Alps
 - (C) Ural Mountains
 - (D) Caucasus

8. Which plateau of the world is known as the 'Roof of the World'?
- (A) Tibetan Plateau (B) Anatolian Plateau
(C) Deccan Plateau (D) Bolivian Plateau
9. The Grand Canyon has been cut by the Colorado river into which plateau?
- (A) Colorado Plateau (B) Great Basin
(C) Ozark Plateau (D) Columbia Plateau
10. Which is the driest desert of the world?
- (A) Gobi (B) Sahara
(C) Atacama (D) Namib
11. The Gobi desert is spread over which of the following? **RAS PRE 2024**
- (A) Kazakhstan and Uzbekistan (B) Iran and Afghanistan
(C) Mongolia and northern China (D) Turkmenistan and Iran
12. The Patagonian desert lies mainly in which country? **RAS PRE 2024**
- (A) Chile (B) Peru
(C) Argentina (D) Bolivia
13. Which of the following is the largest hot desert of the world?
- (A) Sahara (B) Great Victoria
(C) Arabian (D) Kalahari
14. The surface of separation between the crust and the mantle of the earth is known as:
- (A) Gutenberg discontinuity (B) Lehmann discontinuity
(C) Conrad discontinuity (D) Mohorovicic discontinuity
15. The theory of continental drift was propounded by:
- (A) Alfred Wegener (B) Arthur Holmes
(C) W. J. Morgan (D) Harry Hess
16. Which of the following seismic waves cannot pass through the liquid outer core of the earth?
- (A) Primary waves (B) Secondary waves
(C) Love waves (D) Rayleigh waves
17. Which of the following is the highest volcano in the world? **HARD**
- (A) Mauna Loa (B) Mount Etna
(C) Ojos del Salado (D) Cotopaxi
18. The ozone layer of the atmosphere occurs in which layer?
- (A) Troposphere (B) Stratosphere
(C) Mesosphere (D) Thermosphere
19. The local wind 'Chinook', known as the snow-eater, blows in which region?
- (A) Alps of Europe (B) Eastern slopes of the Rocky Mountains
(C) Sahara and the Mediterranean (D) Coast of Argentina
20. The world's highest tidal range is recorded in the:
- (A) Gulf of Mexico (B) Bay of Bengal
(C) Gulf of Carpentaria (D) Bay of Fundy

21. The Great Barrier Reef, the world's largest coral reef system, lies off the coast of:

- (A) Philippines
- (B) Madagascar
- (C) Indonesia
- (D) Australia

22. Consider the following statements about the Circum-Pacific belt:

- I. It is popularly known as the Ring of Fire.
- II. About two-thirds of the world's active volcanoes lie along it.
- III. It consists only of divergent plate boundaries.

Which of the statements given above are correct?

- (A) I, II and III
- (B) I and II only
- (C) II and III only
- (D) I and III only

23. Consider the following statements about Wegener's continental drift theory:

- I. The single supercontinent was named Pangaea.
- II. The ocean surrounding it was named Panthalassa.
- III. Pangaea split into Laurasia in the north and Gondwana in the south.

Which of the statements given above are correct?

- (A) I and III only
- (B) II and III only
- (C) I, II and III
- (D) I and II only

24. Consider the following statements about the Sahara desert:

- I. It is the largest hot desert of the world.
- II. It lies in northern Africa.
- III. The Nile flows through its eastern part.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

25. Consider the following statements about barchans:

- I. They are crescent-shaped sand dunes.
- II. Their horns extend in the direction towards which the wind blows.
- III. They are landforms produced by glacial deposition.

Which of the statements given above are correct?

- (A) I and III only
- (B) II and III only
- (C) I only
- (D) I and II only

26. Consider the following statements:

- I. A fjord is a drowned glacial valley invaded by the sea.
- II. A cirque is an armchair-shaped hollow produced by glacial erosion.
- III. A drumlin is a landform of glacial deposition.

Which of the statements given above are correct?

- (A) I, II and III
- (B) I and II only
- (C) II and III only
- (D) I and III only

27. Consider the following statements about the Tibetan Plateau:

- I. It is the highest plateau in the world.
- II. It lies between the Himalaya and the Kunlun ranges.
- III. It is an example of an intermontane plateau.

Which of the statements given above are correct?

- (A) I and II only
- (B) I, II and III
- (C) II and III only
- (D) I and III only

28. Consider the following statements about tides:

- I. Spring tides occur on full moon and new moon days.
- II. Neap tides occur when the sun and the moon are at right angles with respect to the earth.
- III. Spring tides occur only during the spring season.

Which of the statements given above are correct?

- (A) I, II and III
- (B) II and III only
- (C) I and II only
- (D) I and III only

29. Consider the following statements about the troposphere:

- I. It contains almost all the water vapour and dust of the atmosphere.
- II. Temperature in it falls at an average rate of about 6.5 degrees Celsius per kilometre.
- III. It is thicker over the poles than over the equator.

Which of the statements given above are correct? **HARD**

- (A) I, II and III
- (B) II and III only
- (C) I and III only
- (D) I and II only

30. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Mountain peak) List-II (Continent)

- A. Kilimanjaro 1. North America
- B. Aconcagua 2. Europe
- C. Denali 3. Africa
- D. Elbrus 4. South America

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- (A) A-4, B-3, C-1, D-2
- (B) A-3, B-4, C-1, D-2
- (C) A-3, B-4, C-2, D-1
- (D) A-3, B-1, C-4, D-2

31. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Desert) List-II (Continent / region)

- A. Gobi 1. Australia
 B. Patagonia 2. Southern Africa
 C. Victoria 3. Asia
 D. Kalahari 4. South America

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- (A) A-3, B-4, C-1, D-2
 (B) A-4, B-3, C-1, D-2
 (C) A-1, B-4, C-3, D-2
 (D) A-3, B-1, C-4, D-2

32. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Local wind) List-II (Region)

- A. Mistral 1. West Africa
 B. Harmattan 2. France
 C. Santa Ana 3. Argentina
 D. Zonda 4. California

HARD

- (A) A-2, B-1, C-3, D-4
 (B) A-2, B-1, C-4, D-3
 (C) A-2, B-4, C-1, D-3
 (D) A-3, B-1, C-4, D-2

33. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Ocean current) List-II (Description)

- A. Kuroshio 1. Cold current off north-eastern North America
 B. Benguela 2. Warm current off eastern Japan
 C. Labrador 3. Warm current off south-eastern Africa
 D. Agulhas 4. Cold current off south-western Africa

HARD

- (A) A-2, B-1, C-4, D-3
 (B) A-3, B-4, C-1, D-2
 (C) A-2, B-4, C-1, D-3
 (D) A-2, B-3, C-4, D-1

34. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Plateau) List-II (Country)

- A. Katanga 1. Turkey
 B. Anatolian 2. Democratic Republic of the Congo
 C. Meseta 3. Brazil
 D. Mato Grosso 4. Spain

HARD

- (A) A-2, B-4, C-1, D-3
 (B) A-1, B-2, C-4, D-3
 (C) A-2, B-1, C-4, D-3
 (D) A-2, B-1, C-3, D-4

35. Which one of the following pairs of a mountain peak and its continent is NOT correctly matched? RAS PRE 2016

- (A) Mount Everest - Asia (B) Vinson Massif - Antarctica
 (C) Puncak Jaya - Oceania (D) Mount Kosciuszko - Europe

36. Which one of the following pairs of a desert and its location is NOT correctly matched? RAS PRE 2024

- (A) Thar - India and Pakistan (B) Karakum - Turkmenistan
 (C) Namib - South America (D) Sonoran - North America

37. Which one of the following is NOT a landform produced by glacial action?

- (A) Esker
- (B) Yardang
- (C) Drumlin
- (D) Cirque

38. Which one of the following statements about the layers of the atmosphere is NOT correct?

- (A) The stratosphere is the lowest layer of the atmosphere
- (B) The mesosphere is the layer in which most meteors burn up
- (C) The ionosphere reflects radio waves back to the earth
- (D) The exosphere is the outermost layer of the atmosphere

39. Arrange the following continental highest peaks in descending order of their height:

I. Mount Kilimanjaro

II. Denali

III. Mount Aconcagua

IV. Mount Elbrus RAS PRE 2021

- (A) III, II, IV, I
- (B) II, III, IV, I
- (C) I, III, II, IV
- (D) III, II, I, IV

40. Assertion (A): The Atacama is one of the driest places on the earth.

Reason (R): It lies along a west coast washed by a cold ocean current and in the rain shadow of the Andes.

Select the correct answer using the codes given below: HARD

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is not the correct explanation of A
- (C) A is true but R is false
- (D) A is false but R is true

WG02

Rivers and Lakes of the World

35 questions · 28 fact-sheet points

Must-know facts

1. Nile (about 6,650 km) - longest river of the world and of Africa; rises in the East African highlands via the Kagera; drains into the Mediterranean Sea through a large delta in Egypt.
2. Amazon (about 6,400 km) - second longest but largest in discharge and basin area; rises in the Peruvian Andes, drains into the Atlantic Ocean.
3. Yangtze (about 6,300 km) - longest river of Asia and the longest flowing wholly within one country; drains into the East China Sea.
4. Mississippi-Missouri (about 6,275 km) - longest of North America; drains into the Gulf of Mexico with a bird's-foot delta.
5. Siberian rivers flowing north to the Arctic: Yenisei (5,539 km) to the Kara Sea, Ob-Irtysh (5,410 km) to the Gulf of Ob, Lena (4,400 km) to the Laptev Sea.
6. Huang He / Yellow river (5,464 km) - the 'Sorrow of China' because of its floods; carries a huge load of loess; drains into the Bohai Sea.
7. Volga (about 3,645 km) - longest river of Europe, entirely in Russia, drains into the Caspian Sea. Danube - second longest in Europe, passes through 10 countries and 4 capitals, drains into the Black Sea.
8. Congo (about 4,700 km) - the only major river crossing the Equator twice; the world's deepest river; drains into the Atlantic through an estuary, not a delta.
9. African outfalls: Niger - Gulf of Guinea; Zambezi - Indian Ocean (forms the Victoria Falls); Orange - Atlantic; Limpopo - Indian Ocean.
10. Murray-Darling - longest river system of Australia. Mekong - flows through China, Myanmar, Laos, Thailand, Cambodia and Vietnam into the South China Sea.
11. Tigris and Euphrates unite in southern Iraq to form the Shatt al-Arab, which enters the Persian Gulf.
12. European rivers and cities: Rhine - North Sea (Rotterdam); Rhone - Mediterranean; Elbe - Hamburg; Seine - Paris; Tiber - Rome; Thames - London; Vistula - Warsaw; Spree - Berlin; Danube - Vienna, Bratislava, Budapest, Belgrade.
13. The Blue Nile (from Lake Tana, Ethiopia) and the White Nile meet at Khartoum in Sudan.
14. North American outfalls: Mackenzie - Beaufort Sea (Arctic); Yukon - Bering Sea; Colorado - Gulf of California; St. Lawrence - Atlantic via the Great Lakes.
15. Parana-Rio de la Plata (about 4,880 km) - second longest of South America; Orinoco - Venezuela; Sao Francisco - Brazil.
16. Ganga-Brahmaputra delta (Sundarbans) is the world's largest delta. The Congo and the Amazon discharge through estuaries rather than true deltas.
17. Angel Falls (979 m), Venezuela - world's highest waterfall; plunges from the Auyan-tepui in Canaima National Park into a tributary of the Churun. Tugela Falls, South Africa, is second.
18. Victoria Falls - on the Zambezi, Zambia-Zimbabwe border, called Mosi-oa-Tunya, 'the smoke that thunders'. Niagara - USA-Canada; Iguazu - Argentina-Brazil; Boyoma (Congo) - greatest volume of any waterfall.
19. Suez Canal - opened 1869, about 193 km, joins the Mediterranean (Port Said) to the Red Sea (Suez); a sea-level canal with no locks; nationalised by Egypt in 1956.
20. Panama Canal - opened 1914, about 82 km, joins the Atlantic and the Pacific and works through a system of locks lifting ships to Gatun Lake. Kiel Canal - 1895, about 98 km, North Sea to Baltic Sea.

21. Caspian Sea (about 3.7 lakh sq km) - largest lake on earth, a landlocked salt lake bordered by Russia, Kazakhstan, Turkmenistan, Iran and Azerbaijan.
22. Lake Superior (82,100 sq km) - largest freshwater lake by area; Lake Victoria (about 69,000 sq km) - largest lake of Africa and second largest freshwater lake by area.
23. Lake Baikal - deepest lake (1,642 m) and largest freshwater lake by volume, holding about a fifth of the world's unfrozen surface fresh water; a tectonic rift lake in Siberia. Lake Tanganyika (1,470 m) is second deepest and the world's longest freshwater lake.
24. Lake Titicaca (3,812 m, Peru-Bolivia) - highest navigable lake. Dead Sea shore (about 430 m below sea level) - lowest exposed land surface on earth. Lake Eyre - lowest point and largest salt lake of Australia. Lake Ladoga - largest lake of Europe.
25. Aral Sea (Kazakhstan-Uzbekistan) shrank drastically after the Amu Darya and Syr Darya were diverted for cotton irrigation. Lake Chad is shared by Chad, Niger, Nigeria and Cameroon.
26. Sargasso Sea (North Atlantic) - the only sea without a coastline; it is bounded by ocean currents of the North Atlantic gyre.
27. Straits: Bering (Asia-North America), Gibraltar (Mediterranean-Atlantic; Europe-Africa), Malacca (Sumatra-Malay Peninsula), Hormuz (Persian Gulf-Gulf of Oman), Bab-el-Mandeb (Red Sea-Gulf of Aden), Bosphorus (Black Sea-Sea of Marmara), Magellan, Palk Strait (India-Sri Lanka).
28. Lake types: tectonic (Baikal, Tanganyika, Dead Sea), crater or caldera (Crater Lake, Toba), glacial (Great Lakes), ox-bow (fluvial), lagoon (coastal), artificial (Lake Volta, Lake Nasser).

1. Which is the longest river in the world?

- | | |
|-------------|--------------------------|
| (A) Yangtze | (B) Mississippi-Missouri |
| (C) Amazon | (D) Nile |

2. Which river carries the largest volume of water to the sea?

- | | |
|------------|-------------|
| (A) Nile | (B) Congo |
| (C) Amazon | (D) Yangtze |

3. Which is the longest river of Asia?

- | | |
|------------------|-------------|
| (A) Ob | (B) Yangtze |
| (C) Yellow River | (D) Mekong |

4. The Volga, the longest river of Europe, drains into which water body?

- | | |
|-----------------|----------------|
| (A) Caspian Sea | (B) North Sea |
| (C) Black Sea | (D) Baltic Sea |

5. Which river of the world crosses the Equator twice?

- | | |
|------------|-------------|
| (A) Niger | (B) Zambezi |
| (C) Amazon | (D) Congo |

6. Which river is known as the 'Sorrow of China'?

- | | |
|--------------|--------------|
| (A) Xi Jiang | (B) Huang He |
| (C) Amur | (D) Yangtze |

7. The Victoria Falls are situated on which river?

- | | |
|-------------|------------|
| (A) Zambezi | (B) Orange |
| (C) Congo | (D) Niger |

- 8.** The Angel Falls, the highest waterfall of the world, is located in:
- (A) Venezuela (B) Colombia
(C) Guyana (D) Brazil
- 9.** The Tigris and the Euphrates unite to form which waterway? **HARD**
- (A) Shatt al-Arab (B) Gulf of Aqaba
(C) Karun (D) Orontes
- 10.** The Suez Canal connects the Mediterranean Sea with the:
- (A) Black Sea (B) Caspian Sea
(C) Red Sea (D) Arabian Sea
- 11.** Which of the following canals connects the Atlantic Ocean with the Pacific Ocean?
- (A) Corinth Canal (B) Grand Canal
(C) Kiel Canal (D) Panama Canal
- 12.** Which is the largest lake in the world by surface area?
- (A) Lake Baikal (B) Lake Superior
(C) Caspian Sea (D) Lake Victoria
- 13.** Which is the deepest lake in the world?
- (A) Crater Lake (B) Lake Tanganyika
(C) Caspian Sea (D) Lake Baikal
- 14.** Which is the largest freshwater lake in the world by surface area?
- (A) Lake Superior (B) Lake Michigan
(C) Lake Victoria (D) Lake Huron
- 15.** Lake Titicaca, the highest navigable lake of the world, lies on the border of:
- (A) Peru and Bolivia (B) Ecuador and Colombia
(C) Bolivia and Brazil (D) Chile and Argentina
- 16.** The lowest exposed land surface on the earth is the shore of the:
- (A) Lake Assal (B) Aral Sea
(C) Dead Sea (D) Caspian Sea
- 17.** Which is the only sea in the world that has no coastline? **HARD**
- (A) Andaman Sea (B) Sargasso Sea
(C) Coral Sea (D) Sea of Azov
- 18.** The Bering Strait separates:
- (A) Asia and North America (B) Asia and Australia
(C) North America and South America (D) Europe and Africa
- 19.** The Danube river drains into which sea?
- (A) North Sea (B) Baltic Sea
(C) Black Sea (D) Adriatic Sea

20. Consider the following statements about the Amazon river:

- I. It has the largest drainage basin in the world.
- II. It flows into the Pacific Ocean.
- III. It rises in the Andes of Peru.

Which of the statements given above are correct?

- (A) II and III only
- (B) I, II and III
- (C) I and II only
- (D) I and III only

21. Consider the following statements about Lake Baikal:

- I. It is a tectonic rift lake.
- II. It lies in Siberia in Russia.
- III. It is the deepest lake in the world.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

22. Consider the following statements about the Suez Canal:

- I. It was opened for traffic in 1869.
- II. It is a sea-level canal without locks.
- III. It shortens the sea route between Europe and Asia by avoiding the Cape of Good Hope.

Which of the statements given above are correct?

- (A) I and III only
- (B) I, II and III
- (C) I and II only
- (D) II and III only

23. Consider the following statements about the Nile:

- I. The Blue Nile and the White Nile meet at Khartoum.
- II. It drains into the Red Sea.
- III. It is the longest river of Africa.

Which of the statements given above are correct?

- (A) I and III only
- (B) II and III only
- (C) I, II and III
- (D) I and II only

24. Consider the following statements about the Aral Sea:

- I. It lies between Kazakhstan and Uzbekistan.
- II. It has shrunk drastically because the Amu Darya and the Syr Darya were diverted for irrigation.
- III. It is a freshwater lake fed by the Volga.

Which of the statements given above are correct? **HARD**

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

25. Consider the following statements:

- I. The Ganga-Brahmaputra delta is the largest delta in the world.
- II. The Mississippi has a bird's-foot delta.
- III. The Congo enters the Atlantic Ocean through a large delta.

Which of the statements given above are correct? **HARD**

- (A) I and III only
- (B) I, II and III
- (C) I and II only
- (D) II and III only

26. Match List-I with List-II and select the correct answer using the codes given below:

List-I (River) List-II (Drains into)

- A. Yenisei 1. Gulf of Mexico
- B. Mississippi 2. Kara Sea
- C. Mekong 3. South China Sea
- D. Amur 4. Sea of Okhotsk **HARD**

- (A) A-4, B-1, C-3, D-2
- (B) A-2, B-1, C-3, D-4
- (C) A-2, B-1, C-4, D-3
- (D) A-1, B-2, C-3, D-4

27. Match List-I with List-II and select the correct answer using the codes given below:

List-I (City) List-II (River)

- A. Baghdad 1. Danube
- B. Vienna 2. Tigris
- C. Rome 3. Tiber
- D. Warsaw 4. Vistula

- (A) A-2, B-1, C-4, D-3
- (B) A-2, B-3, C-1, D-4
- (C) A-2, B-1, C-3, D-4
- (D) A-1, B-2, C-3, D-4

28. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Lake) List-II (Country / region)

- A. Lake Titicaca 1. Russia
- B. Lake Eyre 2. Peru and Bolivia
- C. Lake Ladoga 3. Canada
- D. Great Bear Lake 4. Australia

- (A) A-4, B-2, C-1, D-3
- (B) A-2, B-4, C-3, D-1
- (C) A-2, B-4, C-1, D-3
- (D) A-2, B-1, C-4, D-3

29. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Strait) List-II (Connects)

- A. Strait of Hormuz 1. Red Sea and Gulf of Aden
 B. Bab-el-Mandeb 2. Persian Gulf and Gulf of Oman
 C. Strait of Gibraltar 3. Black Sea and Sea of Marmara
 D. Bosphorus 4. Mediterranean Sea and Atlantic Ocean

- (A) A-2, B-1, C-3, D-4
 (B) A-2, B-4, C-1, D-3
 (C) A-2, B-1, C-4, D-3
 (D) A-1, B-2, C-4, D-3

30. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Waterfall) List-II (River)

- A. Niagara Falls 1. Iguazu
 B. Iguazu Falls 2. Niagara
 C. Boyoma Falls 3. Zambezi
 D. Victoria Falls 4. Congo

- (A) A-2, B-3, C-4, D-1
 (B) A-2, B-1, C-4, D-3
 (C) A-2, B-1, C-3, D-4
 (D) A-1, B-2, C-4, D-3

31. Which one of the following pairs of a river and a country through which it flows is NOT correctly matched?

- (A) Irrawaddy - Myanmar (B) Murray - Australia
 (C) Orinoco - Venezuela (D) Tagus - Italy

32. Which one of the following pairs of a city and the river on which it stands is NOT correctly matched?

- (A) Hamburg - Elbe (B) Paris - Rhine
 (C) Khartoum - Nile (D) Cairo - Nile

33. Which one of the following statements about the lakes of the world is NOT correct?

- (A) Lake Victoria is the largest lake of Africa
 (B) Lake Tanganyika is the longest freshwater lake in the world
 (C) Lake Chad is shared by Chad, Niger, Nigeria and Cameroon
 (D) Lake Malawi is the deepest lake in the world

34. Which one of the following pairs relating to the canals of the world is NOT correctly matched?

- (A) Corinth Canal - Greece
 (B) Panama Canal - a sea-level canal without locks
 (C) Grand Canal - China
 (D) Kiel Canal - connects the North Sea with the Baltic Sea

35. Arrange the following rivers in descending order of their length:

- I. Mississippi-Missouri
 II. Nile
 III. Yangtze
 IV. Amazon

- (A) II, IV, I, III
 (B) II, IV, III, I
 (C) IV, II, III, I
 (D) II, III, IV, I

WG03

Natural Vegetation and Biomes

30 questions · 26 fact-sheet points

Must-know facts

1. Equatorial evergreen rainforest ('Selva') - Amazon basin, Congo basin and South-East Asia; multi-storeyed, dense canopy, lianas and epiphytes; no distinct leaf-shedding season.
2. Equatorial hardwoods: mahogany, ebony, rosewood, greenheart, cinchona; scattered occurrence makes commercial exploitation difficult.
3. Tropical deciduous or monsoon forest - teak, sal, sandalwood, bamboo; sheds leaves in the dry season; commercially the most useful forest of the monsoon lands.
4. Savanna (tropical grassland) - tall coarse grass with scattered acacia and baobab; called the 'big game country' for its lion, zebra, giraffe, elephant and wildebeest.
5. Tropical grassland names: Llanos (Orinoco basin, Venezuela), Campos (Brazilian highlands), Sudan/Sahel type (Africa), northern Australian savanna.
6. Temperate grassland-region pairs: Prairie - North America; Pampas - Argentina and Uruguay; Steppe - Eurasia; Veld - South Africa; Downs - Australia; Canterbury - New Zealand; Puszta - Hungary; Manchurian - north-east China.
7. Temperate grasslands are practically treeless, lie in continental interiors with 25-75 cm rainfall, and are the world's granaries and great ranching lands; herdsman of the Pampas are called gauchos.
8. Mediterranean vegetation - hot dry summers and mild wet winters; sclerophyllous plants with thick bark, small waxy leaves and deep roots; citrus, olives, grapes and cork oak; the 'orchard lands' of the world.
9. Mediterranean scrub names: Maquis and Garrigue (Mediterranean basin), Chaparral (California), Matorral (central Chile), Fynbos (Cape region, South Africa), Mallee (south-western Australia).
10. Taiga or boreal coniferous forest - largest land biome, a continuous belt across Siberia, Canada and Scandinavia, confined almost wholly to the Northern Hemisphere.
11. Conifers - pine, fir, spruce, larch, cedar; evergreen, conical, needle-leaved; the world's chief source of softwood, pulp, newsprint and matchwood.
12. Tundra - beyond the tree line, underlain by permafrost; mosses, lichens (reindeer moss) and sedges; growing season only about two months; caribou/reindeer, musk ox, arctic fox, polar bear.
13. Temperate deciduous forest - oak, ash, beech, elm, maple; western Europe, eastern USA and eastern Asia; the most cleared of all forest belts.
14. Desert vegetation - xerophytes such as cactus, acacia, date palm and euphorbia; long roots, thick cuticle, spines in place of leaves, water-storing stems.
15. Mangrove or tidal forest - salt tolerant with stilt roots and pneumatophores; the Sundarbans of the Ganga-Brahmaputra delta is the world's largest mangrove forest.
16. World forest area is about 4.14 billion hectares, roughly 31-32 per cent of the global land area (FAO Global Forest Resources Assessment).
17. More than half of the world's forests are in five countries: Russia (about 832 million ha, one-fifth of the world total), Brazil, Canada, the United States and China.
18. Nearly half of the world's forests are tropical; TROPICAL FORESTS RECORD THE MAXIMUM GLOBAL DEFORESTATION.
19. Global deforestation has slowed from about 17.6 million hectares a year in the 1990s to about 10.9 million hectares a year in the most recent decade.

20. The largest direct driver of tropical deforestation is conversion of forest to agricultural land - shifting cultivation, cattle ranching in the Amazon and oil-palm and soy plantations in South-East Asia and Brazil.
21. The Amazon rainforest, largely in Brazil, is often called the 'lungs of the earth' and holds the world's largest area of tropical rainforest.
22. Biome wildlife: orangutan - Borneo and Sumatra; gorilla, okapi and bonobo - Congo basin; grizzly bear, moose, lynx and wolf - North American coniferous belt; kangaroo - Australian grassland.
23. Sequence of vegetation belts from the Equator poleward: equatorial rainforest, savanna, hot desert, Mediterranean, temperate grassland and deciduous forest, taiga, tundra, ice cap.
24. Vertical (altitudinal) zonation repeats the same sequence up a mountain, ending in alpine pastures above the tree line - the 'alps' of Europe and the bugyals of the Himalaya.
25. Coniferous softwood forests are the world's leading lumbering regions; equatorial hardwood forests are lumbered far less because of mixed stands and poor accessibility.
26. Grassland and vegetation names to memorise: Selva, Llanos, Campos, Pampas, Prairie, Steppe, Veld, Downs, Canterbury, Puszta, Taiga, Tundra, Maquis, Chaparral, Fynbos, Mallee, Matorral.

1. The equatorial evergreen rainforest of the Amazon basin is locally known as:

(A) Llanos	(B) Campos
(C) Taiga	(D) Selva
2. The Prairies are the temperate grasslands of:

(A) Africa	(B) Australia
(C) North America	(D) South America
3. The Pampas grassland is located in:

(A) Brazil	(B) Argentina
(C) Chile	(D) Peru
4. 'Veld' is the name of the temperate grassland of:

(A) Australia	(B) South Africa
(C) Hungary	(D) New Zealand
5. The Canterbury grassland is found in:

(A) Argentina	(B) Australia
(C) New Zealand	(D) South Africa
6. 'Chaparral' is the Mediterranean type of scrub vegetation of:

(A) Southern Africa	(B) Central Chile
(C) California	(D) South-western Australia
7. Which of the following is the largest land biome of the earth?

(A) Tropical rainforest	(B) Taiga or boreal coniferous forest
(C) Savanna	(D) Tundra
8. Reindeer moss and lichens are the characteristic vegetation of which biome?

(A) Tundra	(B) Taiga
(C) Steppe	(D) Savanna

- 9.** At present the maximum deforestation in the world is taking place in which type of forest? RAS PRE 2024
- (A) Mediterranean forests (B) Temperate deciduous forests
(C) Coniferous forests (D) Tropical forests
- 10.** Which country has the largest forest area in the world?
- (A) Canada (B) Russia
(C) China (D) Brazil
- 11.** The tropical grassland of the Orinoco basin in Venezuela is known as:
- (A) Llanos (B) Pampas
(C) Puszta (D) Campos
- 12.** Baobab and acacia trees standing amid tall coarse grasses are typical of which biome?
- (A) Savanna (B) Mediterranean region
(C) Tundra (D) Equatorial rainforest
- 13.** The largest mangrove forest in the world is the:
- (A) Niger delta mangroves (B) Bhitarkanika
(C) Sundarbans (D) Everglades
- 14.** Mahogany, ebony and rosewood are characteristic hardwoods of which forest?
- (A) Coniferous forest (B) Equatorial rainforest
(C) Temperate deciduous forest (D) Mediterranean forest
- 15.** Pine, fir, spruce and larch are the characteristic trees of the:
- (A) Mediterranean scrub (B) Savanna woodland
(C) Monsoon forest (D) Coniferous forest
- 16.** As per the Global Forest Resources Assessment of the FAO, forests cover approximately what proportion of the world's land area?
- (A) About one-tenth (B) About one-half
(C) About one-third (D) About one-fifth
- 17.** Teak, sal and sandalwood are typical trees of which forest type?
- (A) Temperate coniferous forest (B) Tundra vegetation
(C) Tropical evergreen forest (D) Tropical deciduous or monsoon forest
- 18.** Consider the following statements about the equatorial rainforest:
- I. It has a multi-storeyed structure with a dense canopy.
II. Its trees shed their leaves in a definite dry season.
III. Lianas and epiphytes are common in it.
- Which of the statements given above are correct?
- (A) I and III only
(B) II and III only
(C) I, II and III
(D) I and II only

19. Consider the following statements about temperate grasslands:

- I. They are practically treeless.
- II. They are among the world's leading wheat-producing regions.
- III. The Steppes are the temperate grassland of Eurasia.

Which of the statements given above are correct?

- (A) II and III only
- (B) I and III only
- (C) I, II and III
- (D) I and II only

20. Consider the following statements about the taiga:

- I. It occurs as a continuous belt in the Southern Hemisphere.
- II. Its trees are mostly evergreen conifers with needle-shaped leaves.
- III. It is the world's principal source of softwood.

Which of the statements given above are correct? **HARD**

- (A) I and III only
- (B) I, II and III
- (C) I and II only
- (D) II and III only

21. Consider the following statements about deforestation in the world:

- I. Tropical forests record the maximum loss of forest area in the world.
- II. Conversion of forest into agricultural land is the largest direct driver of deforestation.
- III. The annual rate of global deforestation has risen steadily since 1990.

Which of the statements given above are correct? **RAS PRE 2024**

- (A) II and III only
- (B) I and III only
- (C) I, II and III
- (D) I and II only

22. Consider the following statements about Mediterranean vegetation:

- I. The plants are adapted to hot dry summers and mild wet winters.
- II. Leaves are usually small, thick and waxy to check transpiration.
- III. Citrus fruits, olives and grapes are typical of the region.

Which of the statements given above are correct?

- (A) I, II and III
- (B) I and II only
- (C) II and III only
- (D) I and III only

23. Consider the following statements about the tundra biome:

- I. It has a permanently frozen sub-soil called permafrost.
- II. It has a long growing season of more than six months.
- III. Musk ox, arctic fox and polar bear are found in it.

Which of the statements given above are correct? **HARD**

- (A) I and III only
- (B) II and III only
- (C) I, II and III
- (D) I and II only

24. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Temperate grassland) List-II (Region)

- A. Pampas 1. Australia
 B. Downs 2. South Africa
 C. Veld 3. Argentina
 D. Steppe 4. Eurasia
- (A) A-3, B-2, C-1, D-4
 (B) A-1, B-3, C-2, D-4
 (C) A-3, B-1, C-4, D-2
 (D) A-3, B-1, C-2, D-4

25. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Mediterranean scrub) List-II (Region)

- A. Maquis 1. Chile
 B. Fynbos 2. South-western Australia
 C. Mallee 3. Mediterranean basin
 D. Matorral 4. South Africa
- (A) A-3, B-4, C-1, D-2
 (B) A-3, B-4, C-2, D-1
 (C) A-3, B-2, C-4, D-1
 (D) A-1, B-4, C-2, D-3

26. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Biome) List-II (Characteristic animal)

- A. Tundra 1. Kangaroo
 B. Savanna 2. Caribou
 C. Taiga 3. Zebra
 D. Australian grassland 4. Lynx HARD
- (A) A-2, B-4, C-3, D-1
 (B) A-4, B-3, C-2, D-1
 (C) A-2, B-3, C-1, D-4
 (D) A-2, B-3, C-4, D-1

27. Which one of the following pairs of a temperate grassland and its region is NOT correctly matched?

- (A) Canterbury - New Zealand (B) Pampas - South Africa
 (C) Prairie - North America (D) Puszta - Hungary

28. Which one of the following pairs of a vegetation name and its description is NOT correctly matched? HARD

- (A) Taiga - coniferous forest (B) Campos - Mediterranean scrub of Chile
 (C) Selva - equatorial rainforest (D) Llanos - savanna of Venezuela

29. Which one of the following statements regarding the forests of the world is NOT correct? RAS PRE 2024 HARD

- (A) Russia has the largest forest area of any country
 (B) Nearly half of the world's forests are tropical
 (C) Temperate forests are recording the maximum deforestation in the world
 (D) Forests cover about one-third of the world's land area

30. Arrange the following natural vegetation belts in the correct sequence from the Equator towards the North Pole:

I. Tundra

II. Savanna

III. Equatorial rainforest

IV. Taiga

(A) III, II, IV, I

(B) III, II, I, IV

(C) II, III, IV, I

(D) III, IV, II, I

WG04

World Agriculture

35 questions · 27 fact-sheet points

Must-know facts

1. Mediterranean agriculture is confined to five belts: the Mediterranean basin, central California, central Chile, the south-western Cape of South Africa and south-western/southern Australia.
2. Mediterranean lands = 'orchard lands of the world' - citrus, olives, figs, grapes; viticulture (grape growing) is their most distinctive commercial activity.
3. Shifting cultivation local names: Jhum (NE India), Milpa (Mexico/Central America), Ladang (Indonesia-Malaysia), Chena (Sri Lanka), Roca (Brazil), Ray (Vietnam), Taungya (Myanmar), Conuco (Venezuela), Masole (Congo), Humah (Java).
4. Nomadic herders: Bedouin (Arabia-Sahara, camel), Masai (East Africa, cattle), Lapps/Sami (European tundra, reindeer), Chukchi (Siberian tundra), Kirghiz & Kazakh (Central Asian steppe), Fulani (West Africa).
5. Transhumance = seasonal movement of herders between highland summer and lowland winter pastures; Gujjars and Bakarwals in the western Himalayas.
6. Intensive subsistence farming: monsoon Asia, wet paddy dominant, tiny holdings, high yield per hectare but very low output per worker.
7. Extensive commercial grain farming regions: Prairie (USA-Canada), Pampas (Argentina), Steppe (Russia-Ukraine), Veld (South Africa), Downs (Australia), Canterbury (New Zealand).
8. Plantation agriculture: introduced by Europeans in the tropics; single cash crop, large estate, hired labour, heavy capital in processing and transport - rubber (Malaysia), tea (Sri Lanka), coffee (Brazil), cocoa (Ghana), banana (Central America).
9. Kolkhoz = collective farm of the erstwhile USSR (collectivisation from 1929); Sovkhoz = state farm with wage labour; Denmark is the classic case of co-operative farming.
10. Market gardening / truck farming: intensive growing of vegetables, fruit and flowers near big cities; 'factory farming' for stall-fed livestock.
11. Von Thunen's isolated-state model (1826): market gardening and dairying in the innermost ring, forestry next, crop rotation and grain, ranching in the outermost ring.
12. Leading producers (FAO 2023): rice - China then India; wheat - China then India then Russia; maize - USA then China then Brazil; soybean - Brazil then USA then Argentina.
13. Sugarcane - Brazil (about 40 per cent of world output) then India; potato - China then India; groundnut - China then India.
14. Coffee - Brazil is No. 1, Vietnam No. 2 and the largest producer of Robusta; coffee originated in the Ethiopian highlands.
15. Tea - China No. 1, India No. 2, Kenya No. 3, Sri Lanka No. 4; Kenya is the largest exporter of black tea.
16. Cocoa - Cote d'Ivoire No. 1, Ghana No. 2; concentrated on the Gulf of Guinea coast of West Africa.
17. Natural rubber - Thailand No. 1, Indonesia No. 2, Vietnam No. 3; *Hevea brasiliensis* is native to the Amazon basin.
18. Palm oil - Indonesia No. 1 and Malaysia No. 2 together supply about 85 per cent of world output; oil palm is native to West Africa.
19. Olives and olive oil - Spain No. 1, Italy No. 2; jute - India is the largest producer, Bangladesh the largest exporter.
20. Milk - India is the world's largest producer (Operation Flood, Verghese Kurien); New Zealand is the largest exporter of dairy products.

21. Largest exporters: rice - India; wheat - Russia; maize - USA and Brazil; soybean - Brazil; wool - Australia.
22. Fish - China is the largest producer (capture plus aquaculture) and the largest exporter of fishery products.
23. Major fishing grounds: North-West Pacific (world's most productive), North-East Atlantic, Grand Banks off Newfoundland (Labrador Current meets Gulf Stream), Dogger Bank (North Sea), Georges Bank, Peru-Chile coast.
24. The Peruvian anchoveta fishery depends on upwelling caused by the Peru (Humboldt) Current; catches collapse in El Nino years.
25. Norman Borlaug - father of the Green Revolution, Nobel Peace Prize 1970 for dwarf wheat; M. S. Swaminathan led its Indian adaptation from 1966-67.
26. Colour revolutions in India: Green - foodgrains, White - milk, Blue - fish, Yellow - oilseeds, Golden - horticulture/honey, Pink - prawn/onion, Silver - eggs, Red - meat/tomato, Brown - leather/cocoa, Round - potato.
27. FAO was founded on 16 October 1945 with headquarters at Rome; World Food Day is observed on 16 October.

1. Which one of the following regions is NOT associated with the Mediterranean type of agriculture? RAS PRE 2024

- | | |
|--|-----------------------------|
| (A) Central Chile | (B) The Pampas of Argentina |
| (C) The south-western Cape of South Africa | (D) Central California |

2. Mediterranean agriculture is characteristically associated with the cultivation of RAS PRE 2024

- | | |
|--------------------------------------|------------------------------------|
| (A) rubber, cocoa and oil palm | (B) jute, tea and rice |
| (C) citrus fruits, olives and grapes | (D) millets, sorghum and groundnut |

3. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Local name of shifting cultivation) List-II (Region)

- A. Milpa 1. Sri Lanka
 B. Ladang 2. Mexico
 C. Chena 3. Brazil
 D. Roca 4. Indonesia

- (A) A-3, B-4, C-1, D-2
 (B) A-2, B-1, C-4, D-3
 (C) A-4, B-2, C-3, D-1
 (D) A-2, B-4, C-1, D-3

4. Which one of the following countries is the world's largest producer of natural rubber?

- | | |
|---------------|--------------|
| (A) Indonesia | (B) Malaysia |
| (C) Vietnam | (D) Thailand |

5. The world's largest producer of coffee is

- | | |
|--------------|--------------|
| (A) Colombia | (B) Brazil |
| (C) Vietnam | (D) Ethiopia |

6. Which country leads the world in the production of cocoa beans?

- | | |
|-------------------|---------------|
| (A) Cote d'Ivoire | (B) Ghana |
| (C) Nigeria | (D) Indonesia |

- 7.** The world's largest producer of palm oil is
- (A) Nigeria (B) Thailand
(C) Indonesia (D) Malaysia
- 8.** In the erstwhile Soviet Union, the term 'Kolkhoz' referred to a
- (A) state-owned farm on which workers drew wages
(B) collective farm run jointly by its member households
(C) large private estate held by a landlord
(D) co-operative society for the marketing of grain
- 9.** Which country is the world's largest producer of milk?
- (A) India (B) United States of America
(C) Brazil (D) New Zealand
- 10.** Which country is at present the world's largest exporter of rice?
- (A) China (B) Thailand
(C) India (D) Vietnam
- 11.** The world's largest producer of wheat is
- (A) Russia (B) India
(C) United States of America (D) China
- 12.** Which country is the world's largest producer of maize?
- (A) Brazil (B) Argentina
(C) United States of America (D) China
- 13.** Which country is at present the world's largest producer of soybean?
- (A) United States of America (B) Brazil
(C) Argentina (D) China
- 14.** The world's largest producer of sugarcane is
- (A) Brazil (B) India
(C) China (D) Thailand
- 15.** The Grand Banks, one of the world's richest fishing grounds, lie off the coast of
- (A) Norway (B) Peru
(C) Japan (D) Newfoundland
- 16.** Market gardening, also known as truck farming, is best described as the
- (A) cultivation of a single crop on a large estate for export
(B) growing of high-value vegetables, fruits and flowers close to urban markets
(C) rearing of livestock on natural pasture over vast areas
(D) temporary cultivation of food crops on freshly cleared forest land
- 17.** Which country is the world's largest producer of fish, counting both capture fisheries and aquaculture?
- (A) Japan (B) Peru
(C) China (D) Indonesia
- 18.** Which country is at present the world's largest exporter of wheat? **HARD**
- (A) Russia (B) United States of America
(C) Canada (D) Australia

- 19.** Transhumance refers to the
- (A) permanent migration of herders and their families to towns
 - (B) cultivation of fodder crops for stall-fed cattle
 - (C) clearing of forest by burning for temporary cultivation
 - (D) seasonal movement of herders and their livestock between highland and lowland pastures
- 20.** The scientist regarded as the father of the Green Revolution, who received the Nobel Peace Prize in 1970 for developing high-yielding dwarf varieties of wheat, was
- (A) Norman Borlaug
 - (B) M. S. Swaminathan
 - (C) Verghese Kurien
 - (D) Thomas Malthus
- 21.** Which European country, famous for the co-operative organisation of its dairy industry, is popularly described as the 'Dairy of Europe'?
- (A) Switzerland
 - (B) Denmark
 - (C) Netherlands
 - (D) France
- 22.** Consider the following statements about plantation agriculture:
- I. It was introduced in the tropics largely by European settlers.
 - II. It is characterised by the cultivation of a single cash crop on a large estate.
 - III. It requires very little capital investment.
- Which of the statements given above are correct?
- (A) II and III only
 - (B) I and III only
 - (C) I and II only
 - (D) I, II and III
- 23.** Consider the following statements about intensive subsistence agriculture:
- I. It is practised mainly in the densely populated monsoon lands of Asia.
 - II. Holdings are small and the use of human labour is very high.
 - III. Output per hectare is low but output per worker is high.
- Which of the statements given above are correct?
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III
- 24.** Consider the following statements about nomadic herding:
- I. It is practised in the semi-arid and arid lands of Africa and Asia and in the tundra.
 - II. The Masai of East Africa and the Bedouin of Arabia are nomadic herders.
 - III. It is an intensive, market-oriented form of animal rearing.
- Which of the statements given above are correct?
- (A) II and III only
 - (B) I and III only
 - (C) I, II and III
 - (D) I and II only

25. Consider the following statements about von Thunen's model of agricultural land use:

- I. It was put forward in 1826 for an 'isolated state' served by a single central market.
- II. Market gardening and dairying occupy the zone nearest the market.
- III. Extensive ranching occupies the zone farthest from the market.

Which of the statements given above are correct? **HARD**

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

26. Consider the following statements about shifting cultivation:

- I. It involves clearing a patch of forest by cutting and burning the vegetation.
- II. The same plot is cultivated continuously for several decades.

Which of the statements given above is/are correct?

- (A) II only
- (B) I only
- (C) Both I and II
- (D) Neither I nor II

27. Consider the following statements about commercial dairy farming:

- I. It is most highly developed in north-western Europe, north-eastern North America and New Zealand.
- II. Nearness to urban markets and an efficient transport system are essential to it.

Which of the statements given above is/are correct?

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

28. Consider the following statements about the fishery off the coast of Peru:

- I. Its richness is due to the upwelling of cold, nutrient-rich water associated with the Peru (Humboldt) Current.
- II. The anchoveta catch collapses during El Nino years.

Which of the statements given above is/are correct? **HARD**

- (A) Both I and II
- (B) I only
- (C) II only
- (D) Neither I nor II

29. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Crop) List-II (Leading producing country)

- A. Tea 1. Brazil
- B. Coffee 2. Thailand
- C. Cocoa 3. Cote d'Ivoire
- D. Natural rubber 4. China

- (A) A-3, B-1, C-4, D-2
- (B) A-4, B-1, C-3, D-2
- (C) A-4, B-3, C-1, D-2
- (D) A-2, B-1, C-3, D-4

30. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Temperate grassland) List-II (Country)

- A. Prairie 1. Australia
B. Pampas 2. United States of America
C. Veld 3. South Africa
D. Downs 4. Argentina

- (A) A-2, B-3, C-4, D-1
(B) A-1, B-4, C-3, D-2
(C) A-2, B-4, C-1, D-3
(D) A-2, B-4, C-3, D-1

31. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Pastoral community) List-II (Region)

- A. Bedouin 1. Steppes of Central Asia
B. Masai 2. East Africa
C. Lapps (Sami) 3. Tundra of northern Europe
D. Kirghiz 4. Arabian desert **HARD**

- (A) A-4, B-3, C-2, D-1
(B) A-2, B-4, C-3, D-1
(C) A-4, B-2, C-3, D-1
(D) A-1, B-2, C-3, D-4

32. Which one of the following pairs of crop and leading producing country is NOT correctly matched?

- (A) Jute - Bangladesh (B) Olive - Spain
(C) Palm oil - Indonesia (D) Sugarcane - Brazil

33. Which one of the following is NOT a characteristic feature of plantation agriculture?

- (A) Large estates worked by hired labour
(B) Cultivation of a wide variety of food crops for local consumption
(C) Production of a single crop mainly for export
(D) Heavy investment in processing and transport

34. Which one of the following pairs of fishing ground and its location is NOT correctly matched? **HARD**

- (A) Dogger Bank - North Sea
(B) Georges Bank - north-eastern coast of North America
(C) Grand Banks - coast of Norway
(D) Sea of Okhotsk - north-western Pacific

35. Arrange the following countries in descending order of their annual production of tea:

- I. India
II. Kenya
III. China
IV. Sri Lanka

- (A) I, III, IV, II
(B) III, II, I, IV
(C) I, II, III, IV
(D) III, I, II, IV

WG05

World Industry, Transport and Environmental Issues

45 questions · 30 fact-sheet points

Must-know facts

1. Ruhr (Germany) - Europe's foremost coal-based iron and steel region; cities Essen, Dortmund, Duisburg (Rhine inland port); the Saar is the other German coalfield.
2. Lorraine (France) - 'minette' iron ore, Western Europe's largest ore reserve; Nord-Pas-de-Calais is the French coalfield.
3. United Kingdom - Midlands/Black Country (Birmingham, engineering), Lancashire (Manchester, cotton - 'Cottonopolis'), Yorkshire (Leeds-Bradford, wool), Clydeside (Glasgow, shipbuilding), Sheffield (cutlery and special steels).
4. Donbas or Donetsk Basin - Ukraine (coal and steel); Kuzbas (Kuznetsk Basin) and the Urals with Magnitogorsk - Russia; Silesia with Katowice - Poland; Sambre-Meuse - Belgium; Po valley - Italy.
5. Japan - Keihin (Tokyo-Yokohama, largest), Hanshin (Osaka-Kobe), Chukyo (Nagoya), Kitakyushu/Yawata (steel). USA - New England (cotton textiles), Pittsburgh-Lake Erie (steel), Great Lakes with Detroit (automobiles), Silicon Valley in California (electronics and IT). China - Anshan (Manchuria, steel), Shanghai (largest industrial city), Wuhan, Beijing-Tianjin.
6. China makes more than half the world's crude steel; India is second and Japan third. Australia is the largest producer and exporter of iron ore, Brazil (Carajas, Itabira) second.
7. Chile leads world copper output (Chuquibambilla, Escondida) and reserves; Kazakhstan leads uranium; China leads coal and gold; Indonesia leads nickel; Guinea holds the largest bauxite reserves. USA is the largest crude oil producer and Saudi Arabia the largest exporter; OPEC was founded at Baghdad in 1960 with headquarters at Vienna.
8. Trans-Siberian Railway - Moscow to Vladivostok, about 9,289 km, the longest railway line in the world, completed 1916, crossing eight time zones.
9. Trans-Australian Railway (Indian Pacific, Perth-Sydney) carries the world's longest dead-straight stretch of track, about 478 km across the Nullarbor Plain; the Canadian Pacific crosses Canada from Montreal to Vancouver. The Pan-American Highway, the longest motorable road, runs from Alaska to southern Argentina and is broken only by the Darien Gap.
10. Suez Canal - opened 1869, 193 km, Port Said to Suez, links the Mediterranean with the Red Sea, sea-level and lock-free; nationalised by Egypt in 1956.
11. Panama Canal - opened 1914, about 82 km, links the Atlantic with the Pacific through the Gatun, Pedro Miguel and Miraflores locks and Gatun Lake. Kiel Canal joins the North Sea to the Baltic; the Grand Canal of China is the world's longest canal.
12. Shanghai is the world's busiest container port, Singapore second, Rotterdam the largest in Europe; Hartsfield-Jackson Atlanta is the busiest airport by total passengers, Dubai by international passengers, Hong Kong by air cargo.
13. Ozone: Vienna Convention 1985; Montreal Protocol adopted 16 September 1987, in force 1 January 1989 - the first UN treaty with universal ratification. World Ozone Day is 16 September. Ozone lies in the stratosphere and is measured in Dobson Units.
14. Kigali Amendment adopted at Kigali, Rwanda in 2016, in force 1 January 2019 - phase-down of hydrofluorocarbons (HFCs), which are greenhouse gases but not ozone-depleting; India ratified it in 2021.
15. Acid rain - sulphur dioxide and oxides of nitrogen form sulphuric and nitric acid; rain with pH below 5.6; damage to the Taj Mahal ('marble cancer'), the Black Forest and Scandinavian lakes.
16. Stockholm Conference on the Human Environment 1972 created UNEP (headquarters Nairobi) and gave us World Environment Day on 5 June; the UNFCCC was adopted at the Rio Earth Summit in 1992,

- came into force in 1994 and rests on the principle of common but differentiated responsibilities.
17. Kyoto Protocol adopted at COP3 in 1997, in force 16 February 2005, first commitment period 2008-12; flexible mechanisms - Clean Development Mechanism, Joint Implementation, Emissions Trading; Doha Amendment 2012 extended it to 2020.
 18. Kyoto basket of gases - carbon dioxide, methane, nitrous oxide, HFCs, PFCs and sulphur hexafluoride, with nitrogen trifluoride added by the Doha Amendment; sulphur hexafluoride has the highest global warming potential.
 19. Conventional order of contribution to the enhanced greenhouse effect: carbon dioxide (about 60 per cent) > methane (about 20 per cent) > chlorofluorocarbons (about 14 per cent) > nitrous oxide (about 6 per cent).
 20. Paris Agreement adopted at COP21 on 12 December 2015, in force 4 November 2016 - hold warming well below 2 degrees Celsius and pursue 1.5 degrees Celsius, through Nationally Determined Contributions and a global stocktake every five years.
 21. COP hosts and outcomes: COP21 Paris 2015 (Paris Agreement); COP26 Glasgow 2021 (Glasgow Climate Pact, 'phase-down' of unabated coal, India's net-zero 2070 Panchamrit); COP27 Sharm el-Sheikh 2022 (Loss and Damage Fund created); COP28 Dubai 2023 (first Global Stocktake, 'transitioning away from fossil fuels', treble renewables).
 22. COP29 Baku 2024 - New Collective Quantified Goal of at least 300 billion US dollars a year by 2035 within a 1.3 trillion dollar 'Baku Finance Goal'; Article 6 carbon market rules finalised. COP30 Belem 2025 - the 'Global Mutirao' decision, adaptation finance to be trebled by 2035, no fossil-fuel roadmap in the text. COP31 is to be at Antalya, Türkiye, with Australia holding the presidency.
 23. Net-zero target years: India 2070, China 2060, the European Union, United Kingdom, United States and Japan 2050. India's updated NDC - cut the emission intensity of GDP by 45 per cent by 2030 over 2005 and reach 50 per cent non-fossil installed power capacity by 2030.
 24. IPCC was set up in 1988 by the World Meteorological Organization and UNEP; it shared the 2007 Nobel Peace Prize with Al Gore; the AR6 Synthesis Report appeared in 2023.
 25. UNCCD adopted at Paris on 17 June 1994, in force 1996; World Day to Combat Desertification and Drought is 17 June; UNCCD COP16 met at Riyadh in 2024; Land Degradation Neutrality is SDG target 15.3.
 26. Great Green Wall - launched by the African Union in 2007 to create an 8,000 km belt of restored land across the Sahel from Senegal to Djibouti.
 27. Biodiversity hotspots - concept of Norman Myers (1988); 36 hotspots at present; criteria are at least 1,500 endemic vascular plant species and loss of at least 70 per cent of original vegetation; India has four - Himalaya, Indo-Burma, Western Ghats and Sri Lanka, and Sundaland (Nicobar Islands).
 28. Ramsar Convention on Wetlands 1971 (Ramsar, Iran), in force 1975, World Wetlands Day 2 February; CITES 1973 (Washington), in force 1975; CBD 1992 (Rio); Cartagena Protocol on Biosafety 2000; Nagoya Protocol on access and benefit sharing 2010; Kunming-Montreal Global Biodiversity Framework, CBD COP15 Montreal 2022, 23 targets including '30x30'.
 29. Chemicals and waste treaties - Basel 1989 (trans-boundary movement of hazardous wastes), Rotterdam 1998 (prior informed consent), Stockholm 2001 (persistent organic pollutants), Minamata 2013 (mercury).
 30. Plastics - UNEA-5.2 at Nairobi adopted resolution 5/14 'End Plastic Pollution' in March 2022; negotiating sessions at Busan (2024) and Geneva (2025) ended without a treaty. The Great Pacific Garbage Patch lies in the North Pacific gyre between Hawaii and California.

1. Essen, Dortmund and Duisburg are the principal industrial cities of which industrial region? RAS PRE 2016

- | | |
|--------------|--------------|
| (A) The Ruhr | (B) Silesia |
| (C) The Saar | (D) Lorraine |

- 2.** The Kuzbas (Kuznetsk Basin), one of the world's great coal-based industrial regions, lies in
- (A) Ukraine (B) Kazakhstan
(C) Russia (D) Poland
- 3.** Silicon Valley, the world's best known concentration of electronics and information technology industries, lies in
- (A) Texas (B) California
(C) Massachusetts (D) Washington
- 4.** The Keihin industrial region, the largest manufacturing belt of Japan, is centred on
- (A) Osaka and Kobe (B) Nagoya
(C) Kitakyushu (D) Tokyo and Yokohama
- 5.** Which country is the world's largest producer of crude steel?
- (A) Japan (B) India
(C) China (D) United States of America
- 6.** Which country is the world's largest exporter of iron ore?
- (A) Australia (B) Brazil
(C) China (D) South Africa
- 7.** The world's leading producer of copper is
- (A) Peru (B) Chile
(C) China (D) Democratic Republic of the Congo
- 8.** The Trans-Siberian Railway, the longest railway line in the world, runs from Moscow to
- (A) St Petersburg (B) Novosibirsk
(C) Irkutsk (D) Vladivostok
- 9.** The Great Pacific Garbage Patch, the largest known accumulation of floating marine plastic debris, lies between
- (A) Hawaii and California (B) Japan and the Philippines
(C) Australia and New Zealand (D) Peru and Chile
- 10.** Which country is at present the world's largest producer of crude oil?
- (A) Saudi Arabia (B) Russia
(C) United States of America (D) Iran
- 11.** Which one of the following statements about the Suez and Panama Canals is correct?
- (A) Both canals are provided with locks
(B) Neither canal has locks
(C) The Suez Canal has locks while the Panama Canal has none
(D) The Panama Canal has locks while the Suez Canal has none
- 12.** Which is at present the world's busiest airport in terms of total passenger traffic?
- (A) Dubai International (B) Hartsfield-Jackson Atlanta International
(C) Heathrow, London (D) Beijing Capital International
- 13.** Which is at present the world's busiest container port?
- (A) Shanghai (B) Singapore
(C) Rotterdam (D) Busan

- 14.** The Pan-American Highway, generally regarded as the longest motorable road in the world, is broken by a single unbridged stretch of forest and swamp known as the **HARD**
- (A) Culebra Cut (B) Nullarbor Plain
(C) Darien Gap (D) Gran Chaco
- 15.** The Montreal Protocol on Substances that Deplete the Ozone Layer was adopted in the year
- (A) 1985 (B) 1987
(C) 1992 (D) 1997
- 16.** The Kigali Amendment of 2016 to the Montreal Protocol provides for the phase-down of
- (A) chlorofluorocarbons (B) halons
(C) methyl bromide (D) hydrofluorocarbons
- 17.** The International Day for the Preservation of the Ozone Layer is observed every year on
- (A) 5 June (B) 2 February
(C) 16 September (D) 17 June
- 18.** The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 jointly by
- (A) the World Meteorological Organization and the United Nations Environment Programme
(B) the United Nations Development Programme and the World Bank
(C) the Food and Agriculture Organization and UNESCO
(D) the World Health Organization and the World Trade Organization
- 19.** The Paris Agreement on climate change was adopted at which session of the Conference of the Parties to the UNFCCC?
- (A) COP19 (B) COP21
(C) COP24 (D) COP26
- 20.** The long-term temperature goal of the Paris Agreement is to hold the rise in global average temperature to
- (A) below 3 degrees Celsius above pre-industrial levels
(B) below 2.5 degrees Celsius above 1990 levels
(C) exactly 2 degrees Celsius above 2005 levels
(D) well below 2 degrees Celsius above pre-industrial levels while pursuing efforts to limit it to 1.5 degrees Celsius
- 21.** At COP26 held in Glasgow, India announced that it would achieve net-zero emissions by the year
- (A) 2070 (B) 2050
(C) 2060 (D) 2047
- 22.** The decision to establish a dedicated Loss and Damage Fund for countries hit by climate disasters was taken at
- (A) COP25 in Madrid (B) COP26 in Glasgow
(C) COP27 in Sharm el-Sheikh (D) COP28 in Dubai
- 23.** The first Global Stocktake under the Paris Agreement was concluded at
- (A) COP26 in Glasgow (B) COP27 in Sharm el-Sheikh
(C) COP29 in Baku (D) COP28 in Dubai
- 24.** The New Collective Quantified Goal on climate finance agreed at COP29 in Baku commits developed countries to mobilise at least **HARD**
- (A) 100 billion US dollars a year by 2030 (B) 300 billion US dollars a year by 2035
(C) 500 billion US dollars a year by 2040 (D) 1.3 trillion US dollars a year by 2030

25. COP30 of the UNFCCC, held in November 2025, was hosted by which city?

- (A) Belem
- (B) Baku
- (C) Dubai
- (D) Bonn

26. Consider the following statements about the industrial regions of the United States of America:

- I. New England was the earliest centre of cotton textile manufacture in the country.
- II. Detroit in the Great Lakes region is the leading centre of the automobile industry.
- III. Pittsburgh grew as an iron and steel centre on the Appalachian coalfield.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I, II and III
- (D) I and III only

27. Consider the following statements about acid rain:

- I. Sulphur dioxide and oxides of nitrogen are its principal precursors.
- II. It is caused mainly by the release of chlorofluorocarbons from refrigerators and air-conditioners.

Which of the statements given above is/are correct? RAS PRE 2016, 2018

- (A) II only
- (B) I only
- (C) Both I and II
- (D) Neither I nor II

28. Consider the following statements about the Montreal Protocol:

- I. It was preceded by the Vienna Convention for the Protection of the Ozone Layer.
- II. It is the first treaty in the history of the United Nations to be ratified by every member state.
- III. It sets binding targets for the reduction of carbon dioxide emissions.

Which of the statements given above are correct?

- (A) I and III only
- (B) II and III only
- (C) I, II and III
- (D) I and II only

29. Consider the following statements about the Kyoto Protocol:

- I. It was adopted in 1997 and entered into force in 2005.
- II. It placed legally binding emission reduction targets on the developed countries listed in Annexure I.
- III. Clean Development Mechanism, Joint Implementation and Emissions Trading were its three flexible mechanisms.

Which of the statements given above are correct? HARD

- (A) I and II only
- (B) II and III only
- (C) I, II and III
- (D) I and III only

30. Consider the following statements about the Paris Agreement:

- I. Every party is required to prepare and communicate a Nationally Determined Contribution.
- II. It obliges developing countries to accept legally binding absolute cuts in emissions.

Which of the statements given above is/are correct?

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

31. Consider the following statements:

- I. The United Nations Convention to Combat Desertification was adopted in 1994.
- II. The Great Green Wall initiative to check desertification is being implemented across the Sahel region of Africa.
- III. The World Day to Combat Desertification and Drought is observed on 5 June.

Which of the statements given above are correct?

- (A) I and III only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

32. Consider the following statements about biodiversity hotspots:

- I. The concept was first put forward by Norman Myers.
- II. A region must have at least 1,500 endemic species of vascular plants to qualify as a hotspot.
- III. There are at present 36 recognised biodiversity hotspots in the world.

Which of the statements given above are correct? RAS PRE 2016, 2018 HARD

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

33. Consider the following statements about deforestation in the world:

- I. Tropical forests record the highest rate of deforestation among all forest types.
- II. Conversion of forest to cropland and pasture is the single largest driver of deforestation.

Which of the statements given above is/are correct? RAS PRE 2024

- (A) Both I and II
- (B) I only
- (C) II only
- (D) Neither I nor II

34. Which one of the following pairs of industrial region and country is NOT correctly matched? RAS PRE 2013

- | | |
|---------------------|-------------------------------|
| (A) Ruhr - Germany | (B) Midlands - United Kingdom |
| (C) Donbas - Poland | (D) Lorraine - France |

35. Which one of the following is NOT an ozone-depleting substance?

- | | |
|---------------------------|--------------------|
| (A) Chlorofluorocarbon-12 | (B) Carbon dioxide |
| (C) Halon-1211 | (D) Methyl bromide |

36. Which one of the following gases is NOT included in the basket of greenhouse gases regulated under the Kyoto Protocol?

- | | |
|--------------------------|-------------------------|
| (A) Sulphur hexafluoride | (B) Nitrous oxide |
| (C) Perfluorocarbons | (D) Chlorofluorocarbons |

37. Which one of the following pairs of trans-continental railway and country is NOT correctly matched? HARD

- (A) Canadian Pacific Railway - Australia
- (B) Trans-Siberian Railway - Russia
- (C) Union Pacific Railway - United States of America
- (D) Trans-Andean Railway - Chile and Argentina

38. Which one of the following statements about the greenhouse effect is NOT correct?

- (A) Water vapour is the most abundant greenhouse gas in the atmosphere
- (B) Methane has a much higher global warming potential than carbon dioxide over a hundred-year horizon
- (C) Nitrogen is a major greenhouse gas
- (D) The natural greenhouse effect keeps the earth's mean surface temperature well above what it would otherwise be

39. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Industrial region) List-II (Country)

- A. Lorraine 1. Ukraine
- B. Midlands 2. Germany
- C. Ruhr 3. France

D. Donbas 4. United Kingdom RAS PRE 2016, 2023

- (A) A-3, B-2, C-4, D-1
- (B) A-3, B-4, C-2, D-1
- (C) A-2, B-4, C-3, D-1
- (D) A-3, B-4, C-1, D-2

40. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Convention) List-II (Year of adoption)

- A. Ramsar Convention 1. 1992
- B. CITES 2. 2013
- C. Convention on Biological Diversity 3. 1973
- D. Minamata Convention 4. 1971

- (A) A-4, B-1, C-3, D-2
- (B) A-3, B-4, C-1, D-2
- (C) A-4, B-3, C-2, D-1
- (D) A-4, B-3, C-1, D-2

41. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Conference of the Parties) List-II (Host city)

- A. COP26 1. Sharm el-Sheikh
- B. COP27 2. Baku
- C. COP28 3. Glasgow
- D. COP29 4. Dubai

- (A) A-3, B-4, C-1, D-2
- (B) A-1, B-3, C-4, D-2
- (C) A-3, B-1, C-4, D-2
- (D) A-3, B-1, C-2, D-4

42. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Mineral) List-II (Leading producing country)

- A. Copper 1. Kazakhstan
- B. Uranium 2. Chile
- C. Coal 3. China
- D. Nickel 4. Indonesia

- (A) A-2, B-1, C-3, D-4
- (B) A-2, B-4, C-3, D-1
- (C) A-3, B-1, C-2, D-4
- (D) A-2, B-1, C-4, D-3

43. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Industrial city) List-II (Country)

- A. Magnitogorsk 1. Japan
B. Anshan 2. Russia
C. Kitakyushu 3. China
D. Katowice 4. Poland

HARD

- (A) A-2, B-1, C-3, D-4
(B) A-2, B-3, C-1, D-4
(C) A-3, B-2, C-1, D-4
(D) A-2, B-3, C-4, D-1

44. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Convention) List-II (Subject)

- A. Basel Convention 1. Mercury
B. Rotterdam Convention 2. Trans-boundary movement of hazardous wastes
C. Stockholm Convention 3. Prior informed consent for hazardous chemicals
D. Minamata Convention 4. Persistent organic pollutants

HARD

- (A) A-2, B-4, C-3, D-1
(B) A-1, B-3, C-4, D-2
(C) A-2, B-3, C-1, D-4
(D) A-2, B-3, C-4, D-1

45. Arrange the following greenhouse gases in decreasing order of their conventionally estimated contribution to the enhanced greenhouse effect:

- I. Methane
II. Carbon dioxide
III. Nitrous oxide
IV. Chlorofluorocarbons

RAS PRE 2016, 2018

- (A) II, I, IV, III
(B) II, IV, I, III
(C) I, II, III, IV
(D) II, I, III, IV

Answer key & explanations

WG01 – PHYSICAL GEOGRAPHY: MOUNTAINS, PLATEAUS, PLAINS, DESERTS

WG01.1 B	WG01.11 C	WG01.21 D	WG01.31 A
WG01.2 A	WG01.12 C	WG01.22 B	WG01.32 B
WG01.3 D	WG01.13 A	WG01.23 C	WG01.33 C
WG01.4 A	WG01.14 D	WG01.24 D	WG01.34 C
WG01.5 B	WG01.15 A	WG01.25 D	WG01.35 D
WG01.6 D	WG01.16 B	WG01.26 A	WG01.36 C
WG01.7 C	WG01.17 C	WG01.27 B	WG01.37 B
WG01.8 A	WG01.18 B	WG01.28 C	WG01.38 A
WG01.9 A	WG01.19 B	WG01.29 D	WG01.39 D
WG01.10 C	WG01.20 D	WG01.30 B	WG01.40 A

WG02 – RIVERS AND LAKES OF THE WORLD

WG02.1 D	WG02.10 C	WG02.19 C	WG02.28 C
WG02.2 C	WG02.11 D	WG02.20 D	WG02.29 C
WG02.3 B	WG02.12 C	WG02.21 D	WG02.30 B
WG02.4 A	WG02.13 D	WG02.22 B	WG02.31 D
WG02.5 D	WG02.14 A	WG02.23 A	WG02.32 B
WG02.6 B	WG02.15 A	WG02.24 A	WG02.33 D
WG02.7 A	WG02.16 C	WG02.25 C	WG02.34 B
WG02.8 A	WG02.17 B	WG02.26 B	WG02.35 B
WG02.9 A	WG02.18 A	WG02.27 C	

WG03 – NATURAL VEGETATION AND BIOMES

WG03.1 D	WG03.9 D	WG03.17 D	WG03.25 B
WG03.2 C	WG03.10 B	WG03.18 A	WG03.26 D
WG03.3 B	WG03.11 A	WG03.19 C	WG03.27 B
WG03.4 B	WG03.12 A	WG03.20 D	WG03.28 B
WG03.5 C	WG03.13 C	WG03.21 D	WG03.29 C
WG03.6 C	WG03.14 B	WG03.22 A	WG03.30 A
WG03.7 B	WG03.15 D	WG03.23 A	
WG03.8 A	WG03.16 C	WG03.24 D	

WG04 – WORLD AGRICULTURE

WG04.1 B	WG04.10 C	WG04.19 D	WG04.28 A
WG04.2 C	WG04.11 D	WG04.20 A	WG04.29 B
WG04.3 D	WG04.12 C	WG04.21 B	WG04.30 D
WG04.4 D	WG04.13 B	WG04.22 C	WG04.31 C
WG04.5 B	WG04.14 A	WG04.23 A	WG04.32 A
WG04.6 A	WG04.15 D	WG04.24 D	WG04.33 B
WG04.7 C	WG04.16 B	WG04.25 D	WG04.34 C
WG04.8 B	WG04.17 C	WG04.26 B	WG04.35 D
WG04.9 A	WG04.18 A	WG04.27 C	

WG05 – WORLD INDUSTRY, TRANSPORT AND ENVIRONMENTAL ISSUES

WG05.1 A	WG05.3 B	WG05.5 C	WG05.7 B
WG05.2 C	WG05.4 D	WG05.6 A	WG05.8 D

WG05·9 A	WG05·19 B	WG05·29 C	WG05·39 B
WG05·10 C	WG05·20 D	WG05·30 A	WG05·40 D
WG05·11 D	WG05·21 A	WG05·31 B	WG05·41 C
WG05·12 B	WG05·22 C	WG05·32 D	WG05·42 A
WG05·13 A	WG05·23 D	WG05·33 A	WG05·43 B
WG05·14 C	WG05·24 B	WG05·34 C	WG05·44 D
WG05·15 B	WG05·25 A	WG05·35 B	WG05·45 A
WG05·16 D	WG05·26 C	WG05·36 D	
WG05·17 C	WG05·27 B	WG05·37 A	
WG05·18 A	WG05·28 D	WG05·38 C	

WG01 – EXPLANATIONS

- 1. (B) Mount Kilimanjaro** — Mount Kilimanjaro (5,895 m) in Tanzania is Africa's highest peak and a dormant stratovolcano. Mount Kenya (5,199 m) is the continent's second highest.
- 2. (A) Andes** — Aconcagua (about 6,960 m) in Argentina is the highest peak of the Andes and the highest point of the entire Western and Southern Hemispheres.
- 3. (D) Caucasus** — Mount Elbrus (5,642 m) in Russia lies in the Caucasus range and is the highest point of Europe. Mont Blanc (4,808 m) in the Alps is the highest peak of Western Europe.
- 4. (A) Antarctica** — Vinson Massif (4,892 m) in the Sentinel Range of the Ellsworth Mountains is the highest peak of Antarctica.
- 5. (B) United States of America** — Denali (6,190 m), formerly called Mount McKinley, stands in the Alaska Range in the United States. Its officially accepted height was revised to 6,190 m in 2015.
- 6. (D) Andes** — The Andes run about 7,000 km along the western edge of South America through seven countries. The Mid-Oceanic Ridge (about 65,000 km) is longer but lies almost entirely under the sea.
- 7. (C) Ural Mountains** — The Ural Mountains, running roughly north to south across Russia, form the conventional dividing line between Europe and Asia.
- 8. (A) Tibetan Plateau** — The Tibetan Plateau, lying between the Himalaya and the Kunlun ranges, is the world's highest and largest plateau, with an average elevation above 4,500 m.
- 9. (A) Colorado Plateau** — The Grand Canyon is carved into the Colorado Plateau of the south-western United States, the classic example of a plateau dissected by river erosion.
- 10. (C) Atacama** — The Atacama desert of northern Chile is the driest non-polar desert, kept arid by the cold Peru (Humboldt) current and by the rain shadow of the Andes.
- 11. (C) Mongolia and northern China** — The Gobi, a cold-winter desert of Asia, extends across southern Mongolia and northern China. The Karakum lies in Turkmenistan and the Kyzylkum in Uzbekistan and Kazakhstan.
- 12. (C) Argentina** — The Patagonian desert (about 6.73 lakh sq km) of southern Argentina is the largest desert of the Americas. It is a cold rain-shadow desert lying east of the Andes.
- 13. (A) Sahara** — The Sahara of northern Africa, over 8.5 million sq km in extent, is the world's largest hot desert. The Antarctic and Arctic polar deserts are larger overall.
- 14. (D) Mohorovicic discontinuity** — The Mohorovicic (Moho) discontinuity separates the crust from the mantle, while the Gutenberg discontinuity separates the mantle from the core.
- 15. (A) Alfred Wegener** — Alfred Wegener put forward the continental drift theory in 1912, proposing a single supercontinent Pangaea surrounded by the ocean Panthalassa. Harry Hess later proposed sea-floor spreading.
- 16. (B) Secondary waves** — Secondary (S) waves are transverse and travel only through solids. Their absence beyond about 103 degrees creates the shadow zone and proves that the outer core is liquid.
- 17. (C) Ojos del Salado** — Nevado Ojos del Salado, on the Chile-Argentina border in the Andes, is the world's highest volcano. Mauna Loa in Hawaii is the largest active volcano by volume.
- 18. (B) Stratosphere** — The ozone layer lies within the stratosphere, roughly 15 to 35 km above the surface, and absorbs harmful ultraviolet radiation from the sun.

- 19. (B) Eastern slopes of the Rocky Mountains** — The Chinook is a warm dry descending wind on the eastern slopes of the Rockies in North America; it melts snow rapidly, hence 'snow-eater'. Its European counterpart is the Foehn of the Alps.
- 20. (D) Bay of Fundy** — The Bay of Fundy in eastern Canada records the world's highest tidal range, exceeding 15 metres, because of its funnel shape and tidal resonance.
- 21. (D) Australia** — The Great Barrier Reef lies in the Coral Sea off the coast of Queensland in north-eastern Australia and is a UNESCO World Heritage Site.
- 22. (B) I and II only** — The Circum-Pacific belt or Ring of Fire is the world's greatest earthquake and volcano belt, but it is dominated by convergent boundaries and subduction zones, not by divergent boundaries.
- 23. (C) I, II and III** — All three statements are correct. The narrow sea separating Laurasia from Gondwana was the Tethys, from whose sediments the Himalaya were later folded up.
- 24. (D) I, II and III** — All three statements are correct. The Sahara spreads over more than 8.5 million sq km of northern Africa and is crossed in the east by the Nile, which sustains Egypt's settled agriculture.
- 25. (D) I and II only** — Barchans are crescentic dunes formed by wind (aeolian) deposition, with horns pointing downwind; they have nothing to do with glaciers. Dunes aligned parallel to the wind are called seifs.
- 26. (A) I, II and III** — All three statements are correct. Norway, Chile, New Zealand and Greenland have the world's best developed fjord coasts.
- 27. (B) I, II and III** — All three statements are correct. An intermontane plateau is one enclosed by mountain ranges; the Tibetan, Bolivian and Anatolian plateaus are of this type.
- 28. (C) I and II only** — Spring tides are the highest tides and occur when the sun, earth and moon are in a line (syzygy); they have nothing to do with the spring season. Neap tides occur at quadrature.
- 29. (D) I and II only** — The troposphere is about 18 km thick over the equator but only about 8 km over the poles, so the third statement is wrong. All weather phenomena occur in this lowest layer.
- 30. (B) A-3, B-4, C-1, D-2** — Kilimanjaro is the highest peak of Africa, Aconcagua of South America, Denali of North America and Elbrus of Europe.
- 31. (A) A-3, B-4, C-1, D-2** — The Gobi lies in Asia (Mongolia and China), Patagonia in South America (Argentina), the Great Victoria in Australia and the Kalahari in southern Africa.
- 32. (B) A-2, B-1, C-4, D-3** — The Mistral is a cold wind of southern France, the Harmattan a dry dusty wind of West Africa, the Santa Ana a hot dry wind of southern California and the Zonda a warm descending wind of Argentina.
- 33. (C) A-2, B-4, C-1, D-3** — The Kuroshio is a warm Pacific current off Japan, the Benguela a cold current off south-western Africa, the Labrador a cold current off north-eastern North America and the Agulhas a warm current off south-eastern Africa.
- 34. (C) A-2, B-1, C-4, D-3** — The Katanga (Shaba) plateau of the DR Congo is a great copper belt, the Anatolian plateau lies in Turkey, the Meseta is the central plateau of Spain and the Mato Grosso plateau is in Brazil.
- 35. (D) Mount Kosciuszko - Europe** — Mount Kosciuszko (2,228 m) is the highest peak of mainland Australia, not of Europe. The highest peak of Europe is Mount Elbrus in the Caucasus.
- 36. (C) Namib - South America** — The Namib is a coastal desert of south-western Africa, mainly in Namibia, and is regarded as the world's oldest desert. The chief South American deserts are the Atacama and the Patagonian.
- 37. (B) Yardang** — A yardang is a wind-eroded ridge of a desert region. Cirques are landforms of glacial erosion, while eskers and drumlins are of glacial deposition.
- 38. (A) The stratosphere is the lowest layer of the atmosphere** — The troposphere, not the stratosphere, is the lowest layer of the atmosphere. The stratosphere lies above the tropopause and contains the ozone layer.
- 39. (D) III, II, I, IV** — Aconcagua (about 6,960 m) > Denali (6,190 m) > Kilimanjaro (5,895 m) > Elbrus (5,642 m). Mount Everest (8,848.86 m) tops the list of continental highest peaks.
- 40. (A) Both A and R are true and R is the correct explanation of A** — The cold Peru (Humboldt) current chills the air over the coast and suppresses convection, while the Andes block moisture coming from the Amazon side; together they keep the Atacama extremely arid.

WG02 – EXPLANATIONS

1. **(D) Nile** — The Nile, about 6,650 km long, is conventionally reckoned the world's longest river. It rises in the East African highlands and drains into the Mediterranean Sea through a large delta in Egypt.
2. **(C) Amazon** — The Amazon has by far the greatest discharge and the largest drainage basin of any river, accounting for roughly a fifth of the world's total river flow into the oceans.
3. **(B) Yangtze** — The Yangtze, about 6,300 km long, is Asia's longest river and the longest river flowing entirely within a single country. It drains into the East China Sea.
4. **(A) Caspian Sea** — The Volga, about 3,645 km long, flows entirely within Russia and empties into the Caspian Sea, an inland basin with no outlet to the ocean.
5. **(D) Congo** — The Congo crosses the Equator twice in its great arc through central Africa, which gives it a fairly even flow throughout the year. It is also the world's deepest river.
6. **(B) Huang He** — The Huang He or Yellow River earned this name from its disastrous floods. Its yellow colour and raised bed come from the enormous load of loess that it carries.
7. **(A) Zambezi** — The Victoria Falls, locally called Mosi-oa-Tunya or 'the smoke that thunders', lie on the Zambezi on the Zambia-Zimbabwe border.
8. **(A) Venezuela** — The Angel Falls (979 m) plunges from the Auyan-tepui in Canaima National Park, Venezuela, into a tributary of the Churun river. Tugela Falls in South Africa is the second highest.
9. **(A) Shatt al-Arab** — The Tigris and the Euphrates join in southern Iraq to form the Shatt al-Arab, which then flows into the Persian Gulf.
10. **(C) Red Sea** — Opened in 1869 and about 193 km long, the Suez Canal joins the Mediterranean at Port Said with the Red Sea at Suez. Being a sea-level canal, it has no locks.
11. **(D) Panama Canal** — The Panama Canal, opened in 1914 and about 82 km long, links the Atlantic with the Pacific and works through a system of locks that lift ships up to Gatun Lake.
12. **(C) Caspian Sea** — The Caspian Sea, bordered by Russia, Kazakhstan, Turkmenistan, Iran and Azerbaijan, is a landlocked salt lake of roughly 3.7 lakh sq km, the largest lake on earth.
13. **(D) Lake Baikal** — Lake Baikal in Siberia reaches a depth of 1,642 m and holds about a fifth of the world's unfrozen surface fresh water. Lake Tanganyika (1,470 m) is the second deepest.
14. **(A) Lake Superior** — Lake Superior, about 82,100 sq km in area and one of the Great Lakes of North America, is the largest freshwater lake by surface area. Lake Victoria in Africa is second.
15. **(A) Peru and Bolivia** — Lake Titicaca lies at about 3,812 m on the Altiplano of the Andes between Peru and Bolivia.
16. **(C) Dead Sea** — The shore of the Dead Sea, between Israel and Jordan, lies about 430 m below sea level. Its extreme salinity makes bathers float easily.
17. **(B) Sargasso Sea** — The Sargasso Sea in the North Atlantic is bounded not by land but by the ocean currents of the North Atlantic gyre, and is named after its floating Sargassum weed.
18. **(A) Asia and North America** — The Bering Strait separates Russia from Alaska and links the Bering Sea with the Arctic Ocean. The Strait of Gibraltar separates Europe from Africa.
19. **(C) Black Sea** — The Danube, Europe's second longest river, rises in the Black Forest of Germany, passes through ten countries and four national capitals, and empties into the Black Sea.
20. **(D) I and III only** — The Amazon rises in the Peruvian Andes and flows eastward across Brazil into the Atlantic Ocean, not the Pacific. Its basin is the largest in the world.
21. **(D) I, II and III** — All three statements are correct. Baikal occupies a continental rift valley, is more than 25 million years old and is a UNESCO World Heritage Site.
22. **(B) I, II and III** — All three statements are correct. Egypt nationalised the canal in 1956, an act that led to the Suez crisis of that year.
23. **(A) I and III only** — The Nile drains into the Mediterranean Sea, not the Red Sea. The Blue Nile, rising from Lake Tana in Ethiopia, joins the White Nile at Khartoum in Sudan.

- 24. (A) I and II only** — The Aral Sea is a saline lake fed by the Amu Darya and the Syr Darya, not by the Volga. Heavy diversion of these rivers for cotton irrigation caused one of the worst man-made ecological disasters of the last century.
- 25. (C) I and II only** — The Congo discharges through a deep estuary and submarine canyon rather than a delta, because strong currents and its steep gradient prevent deposition at the mouth.
- 26. (B) A-2, B-1, C-3, D-4** — The Yenisei drains into the Kara Sea, the Mississippi into the Gulf of Mexico, the Mekong into the South China Sea and the Amur into the Sea of Okhotsk.
- 27. (C) A-2, B-1, C-3, D-4** — Baghdad stands on the Tigris, Vienna on the Danube, Rome on the Tiber and Warsaw on the Vistula.
- 28. (C) A-2, B-4, C-1, D-3** — Lake Eyre in South Australia is that continent's largest salt lake and its lowest point, while Lake Ladoga in Russia is the largest lake of Europe.
- 29. (C) A-2, B-1, C-4, D-3** — About a fifth of the world's oil trade passes through the Strait of Hormuz, and the Bab-el-Mandeb is the southern gateway to the Red Sea and the Suez route.
- 30. (B) A-2, B-1, C-4, D-3** — The Boyoma (formerly Stanley) Falls on the Congo carry the greatest volume of water of any waterfall in the world.
- 31. (D) Tagus - Italy** — The Tagus is the longest river of the Iberian Peninsula, flowing through Spain and Portugal and reaching the Atlantic at Lisbon. The chief river of Italy is the Po.
- 32. (B) Paris - Rhine** — Paris stands on the Seine. The Rhine passes Basel, Cologne and Rotterdam before entering the North Sea.
- 33. (D) Lake Malawi is the deepest lake in the world** — Lake Baikal is the deepest lake in the world at 1,642 m, followed by Lake Tanganyika at 1,470 m. Lake Malawi is far shallower than either.
- 34. (B) Panama Canal - a sea-level canal without locks** — The Panama Canal works through a system of locks that raise ships to Gatun Lake and lower them again. It is the Suez Canal that is a sea-level canal without locks.
- 35. (B) II, IV, III, I** — Nile (about 6,650 km) > Amazon (about 6,400 km) > Yangtze (about 6,300 km) > Mississippi-Missouri (about 6,275 km).

WG03 – EXPLANATIONS

- 1. (D) Selva** — 'Selva' is the name given to the dense equatorial rainforest of the Amazon. Llanos and Campos are savanna grasslands of the Orinoco basin and the Brazilian highlands respectively.
- 2. (C) North America** — The Prairies stretch across the central plains of the United States and Canada and form one of the world's great wheat belts.
- 3. (B) Argentina** — The Pampas of Argentina and Uruguay are famous for wheat, alfalfa and cattle ranching; the herdsmen of the region are called gauchos.
- 4. (B) South Africa** — The Veld, divided into high veld, middle veld and low veld, lies on the plateau of South Africa and supports large-scale maize cultivation and sheep and cattle rearing.
- 5. (C) New Zealand** — The Canterbury Plains on the eastern side of South Island, New Zealand, are famous for sheep rearing and for lamb and wool exports.
- 6. (C) California** — Chaparral is the Californian name for Mediterranean sclerophyll scrub. Its counterparts are maquis in the Mediterranean basin, matorral in Chile, fynbos in South Africa and mallee in Australia.
- 7. (B) Taiga or boreal coniferous forest** — The taiga sweeps in a broad belt across Siberia, Canada and Scandinavia and is the largest land biome. It supplies the world's chief stocks of softwood, pulp and newsprint.
- 8. (A) Tundra** — The tundra, lying beyond the tree line and underlain by permafrost, supports only mosses, lichens and sedges. Reindeer or caribou graze on reindeer moss.
- 9. (D) Tropical forests** — Tropical forests, particularly in the Amazon, the Congo basin and South-East Asia, account for the bulk of the world's forest loss, driven by clearing for agriculture, ranching and plantations.
- 10. (B) Russia** — Russia has about 832 million hectares of forest, roughly one-fifth of the world's total, followed by Brazil, Canada, the United States and China.

- 11. (A) Llanos** — The Llanos are the savanna of the Orinoco basin, while the Campos are the savanna of the Brazilian highlands. Both are tropical, not temperate, grasslands.
- 12. (A) Savanna** — The savanna or tropical grassland has tall grass with scattered drought-resistant trees such as acacia and baobab. It is called the 'big game country' for its lions, zebras, giraffes and elephants.
- 13. (C) Sundarbans** — The Sundarbans, in the Ganga-Brahmaputra delta of India and Bangladesh, is the world's largest mangrove forest and the home of the Royal Bengal tiger.
- 14. (B) Equatorial rainforest** — Equatorial rainforests yield heavy, durable hardwoods such as mahogany, ebony, rosewood and greenheart. Their scattered occurrence in mixed stands makes commercial exploitation difficult.
- 15. (D) Coniferous forest** — These conifers have needle-shaped leaves and a conical shape that sheds snow, and they supply the softwood used for paper pulp, matchwood and furniture.
- 16. (C) About one-third** — The world's forests extend over about 4.14 billion hectares, roughly 31 to 32 per cent of the global land area. Just five countries hold more than half of this area.
- 17. (D) Tropical deciduous or monsoon forest** — Tropical deciduous or monsoon forests shed their leaves in the dry season to conserve water. They are commercially the most useful forests of the monsoon lands.
- 18. (A) I and III only** — Rainfall and temperature are high throughout the year in the equatorial belt, so there is no distinct leaf-shedding season and the forest remains evergreen.
- 19. (C) I, II and III** — All three statements are correct. Temperate grasslands lie in continental interiors with 25 to 75 cm of rainfall and are the world's granaries as well as great ranching lands.
- 20. (D) II and III only** — There is very little land in the corresponding latitudes of the Southern Hemisphere, so the taiga is confined almost entirely to the Northern Hemisphere.
- 21. (D) I and II only** — The rate of global deforestation has in fact declined, from about 17.6 million hectares a year in the 1990s to about 10.9 million hectares a year in the most recent decade, though tropical forests still bear the greatest loss.
- 22. (A) I, II and III** — All three statements are correct. Mediterranean lands lie on the western margins of continents between about 30 and 40 degrees latitude and are often called the orchard lands of the world.
- 23. (A) I and III only** — The tundra's growing season lasts only about two months of the brief Arctic summer, when the surface thaws and mosses, lichens and sedges appear.
- 24. (D) A-3, B-1, C-2, D-4** — The Pampas are in Argentina, the Downs in Australia, the Veld in South Africa and the Steppes in Eurasia. Canterbury in New Zealand and Puszta in Hungary complete the list.
- 25. (B) A-3, B-4, C-2, D-1** — The five Mediterranean climate regions are the Mediterranean basin (maquis and garrigue), California (chaparral), central Chile (matorral), the Cape region of South Africa (fynbos) and south-western Australia (mallee).
- 26. (D) A-2, B-3, C-4, D-1** — Caribou or reindeer graze the tundra, zebra the African savanna, lynx the coniferous taiga and kangaroos the grasslands of Australia.
- 27. (B) Pampas - South Africa** — The Pampas are the temperate grassland of Argentina and Uruguay in South America. The temperate grassland of South Africa is the Veld.
- 28. (B) Campos - Mediterranean scrub of Chile** — Campos is the tropical savanna grassland of the Brazilian highlands. The Mediterranean scrub of central Chile is called matorral.
- 29. (C) Temperate forests are recording the maximum deforestation in the world** — Tropical forests, not temperate forests, record the maximum deforestation. Temperate and boreal forest areas have in fact been broadly stable or expanding in recent decades.
- 30. (A) III, II, IV, I** — Moving poleward from the Equator the sequence is equatorial rainforest, savanna, hot desert, Mediterranean, temperate grassland and deciduous forest, taiga and finally tundra.

WG04 — EXPLANATIONS

1. **(B) The Pampas of Argentina** — Mediterranean agriculture is confined to the five Mediterranean-climate belts - the Mediterranean basin, central California, central Chile, the south-western Cape of South Africa and south-western/southern Australia. The Pampas is a temperate grassland of extensive commercial grain farming and cattle ranching.

2. **(C) citrus fruits, olives and grapes** — Mediterranean lands specialise in citrus fruits, olives, figs and grapes and are called the 'orchard lands of the world'. Viticulture, or grape growing for wine, is their most distinctive commercial activity.

3. **(D) A-2, B-4, C-1, D-3** — Shifting cultivation is called Milpa in Mexico and Central America, Ladang in Indonesia and Malaysia, Chena in Sri Lanka and Roca in Brazil. In north-eastern India it is known as Jhum.

4. **(D) Thailand** — Thailand has been the world's leading producer of natural rubber for decades, ahead of Indonesia and Vietnam. The rubber tree, *Hevea brasiliensis*, is native to the Amazon basin.

5. **(B) Brazil** — Brazil has been the world's largest coffee producer for over a century and a half; Vietnam is second and is the largest producer of Robusta coffee. The plant itself originated in the Ethiopian highlands.

6. **(A) Cote d'Ivoire** — Cote d'Ivoire produces roughly two-fifths of the world's cocoa, ahead of Ghana. Cocoa cultivation is concentrated along the Gulf of Guinea coast of West Africa.

7. **(C) Indonesia** — Indonesia and Malaysia together account for about 85 per cent of world palm oil output, with Indonesia far ahead. The oil palm, *Elaeis guineensis*, is native to West Africa.

8. **(B) collective farm run jointly by its member households** — The Kolkhoz was the collective farm formed by pooling the land, implements and livestock of member households; the Sovkhoz was the state farm worked by wage labour. Collectivisation in the USSR began in 1929.

9. **(A) India** — India has been the world's largest milk producer since the 1990s, an outcome of Operation Flood led by Verghese Kurien. New Zealand, though a modest producer, is the largest exporter of dairy products.

10. **(C) India** — India is the world's largest rice exporter, followed by Thailand and Vietnam. China, though the biggest producer of rice, consumes almost its entire output at home.

11. **(D) China** — China is the largest wheat producer, followed by India and Russia; the three together account for well over 40 per cent of world output. Wheat is a rabi crop of the temperate and sub-tropical lands.

12. **(C) United States of America** — The United States produces roughly a third of the world's maize, ahead of China and Brazil. The American Corn Belt centres on Iowa and Illinois in the Middle West.

13. **(B) Brazil** — Brazil overtook the United States as the largest soybean producer around 2019-20 and has held first place since; Argentina is third and leads in exports of soybean oil and meal.

14. **(A) Brazil** — Brazil grows about 40 per cent of the world's sugarcane, with India second. A large share of the Brazilian crop is converted into ethanol rather than sugar.

15. **(D) Newfoundland** — The Grand Banks lie off Newfoundland in Canada, where the cold Labrador Current meets the warm Gulf Stream and the resulting plankton bloom supports huge cod stocks. The Dogger Bank is the corresponding shallow bank in the North Sea.

16. **(B) growing of high-value vegetables, fruits and flowers close to urban markets** — Market gardening is an intensive, capital- and labour-intensive form of cultivation of vegetables, fruits and flowers for nearby city markets. In the United States it is called truck farming because the produce is trucked in daily.

17. **(C) China** — China accounts for well over a third of world fish production and is also the largest exporter of fish and fishery products. Peru's output, by contrast, is dominated by a single species, the anchoveta.

18. **(A) Russia** — Russia has been the world's largest wheat exporter since the mid-2010s, shipping mainly through its Black Sea ports. The European Union, the United States, Canada and Australia follow.

19. **(D) seasonal movement of herders and their livestock between highland and lowland pastures** — Transhumance is the regular seasonal movement of pastoralists with their animals between summer pastures on high ground and winter pastures in the valleys - as practised by the Gujjars and Bakarwals of the western Himalayas.

20. **(A) Norman Borlaug** — Norman Borlaug developed semi-dwarf, rust-resistant wheat in Mexico and won the 1970 Nobel Peace Prize. M. S. Swaminathan led the adaptation of these varieties in India from 1966-67.

21. **(B) Denmark** — Denmark's dairying is organised almost wholly through producers' co-operatives, which makes the country the standard textbook example both of co-operative farming and of commercial dairy farming.

- 22. (C) I and II only** — Plantations are large, single-crop, export-oriented estates set up by Europeans in tropical colonies. They are highly capital-intensive, needing heavy investment in processing plants, transport and management, so statement III is wrong.
- 23. (A) I and II only** — Intensive subsistence farming in monsoon Asia yields a high output per hectare but a very low output per worker, because a large labour force works tiny holdings. Statement III reverses this relationship.
- 24. (D) I and II only** — Nomadic herding is an extensive, largely subsistence activity carried on in harsh environments - the Sahara-Arabia belt, Central Asia and the tundra. Commercial livestock ranching is its market-oriented counterpart.
- 25. (D) I, II and III** — Von Thunen's concentric-ring model (Der isolierte Staat, 1826) places perishable and bulky produce - market gardening and dairying - closest to the market and ranching in the outermost ring, because transport cost rises with distance.
- 26. (B) I only** — Shifting or slash-and-burn cultivation clears and burns a forest patch, crops it for two or three years until fertility falls and then abandons it for a long fallow. It is called Jhum in north-eastern India.
- 27. (C) Both I and II** — Milk is bulky and perishable, so dairy belts develop close to large urban markets and depend on refrigerated transport. Denmark, the Netherlands, north-eastern USA, Canada, New Zealand and south-eastern Australia are the leading dairy regions.
- 28. (A) Both I and II** — Upwelling along the Peruvian coast brings nutrients to the surface and sustains the world's largest single-species fishery, the anchoveta. During El Nino the intrusion of warm water suppresses upwelling and the catch collapses.
- 29. (B) A-4, B-1, C-3, D-2** — China leads in tea, Brazil in coffee, Cote d'Ivoire in cocoa and Thailand in natural rubber. India is second in tea and Vietnam second in coffee.
- 30. (D) A-2, B-4, C-3, D-1** — The Prairies (USA and Canada), Pampas (Argentina), Veld (South Africa) and Downs (Australia) are the great temperate grasslands where extensive commercial grain farming and livestock ranching are carried on.
- 31. (C) A-4, B-2, C-3, D-1** — The Bedouin herd camels and goats in the Arabian desert, the Masai cattle in East Africa, the Lapps or Sami reindeer in the European tundra and the Kirghiz sheep and horses on the Central Asian steppes.
- 32. (A) Jute - Bangladesh** — India, not Bangladesh, is the world's largest producer of jute; Bangladesh is the largest exporter of jute and jute goods. Spain leads in olives and olive oil, Indonesia in palm oil and Brazil in sugarcane.
- 33. (B) Cultivation of a wide variety of food crops for local consumption** — Plantations are monocultures grown for distant export markets, not diversified food farms. Rubber in Malaysia, tea in Sri Lanka, coffee in Brazil and cocoa in Ghana are the classic examples.
- 34. (C) Grand Banks - coast of Norway** — The Grand Banks lie off Newfoundland in Canada, not off Norway. The north-west Pacific, which includes the Sea of Okhotsk and the seas around Japan, is the world's most productive fishing region.
- 35. (D) III, I, II, IV** — China is the world's largest tea producer, followed by India, Kenya and Sri Lanka. Kenya is, however, the world's largest exporter of black tea.

WG05 — EXPLANATIONS

- 1. (A) The Ruhr** — These cities lie in the Ruhr basin of western Germany, Europe's foremost coal-based iron and steel region, served by the Rhine waterway and the great inland port of Duisburg.
- 2. (C) Russia** — The Kuznetsk basin in western Siberia is Russia's largest coalfield and the base of the Kuzbas iron and steel region. Magnitogorsk in the Urals is the other great Russian steel centre.
- 3. (B) California** — Silicon Valley occupies the Santa Clara Valley of California around San Jose. The Route 128 belt near Boston in New England is the other major American high-technology cluster.
- 4. (D) Tokyo and Yokohama** — Keihin (Tokyo-Yokohama) on the Kanto plain is Japan's leading industrial region. Hanshin is Osaka-Kobe, Chukyo is Nagoya and Kitakyushu on Kyushu is the old iron and steel belt.
- 5. (C) China** — China produces more than half of the world's crude steel. India is second, having overtaken Japan in 2018, and Japan third.

- 6. (A) Australia** — Australia, drawing mainly on the Pilbara ranges of Western Australia, is both the largest producer and the largest exporter of iron ore; Brazil, with Carajas and Itabira, is second.
- 7. (B) Chile** — Chile, with the Chuquibambilla and Escondida mines in the Atacama region, leads world copper production and also holds the largest reserves; Peru is second.
- 8. (D) Vladivostok** — The Trans-Siberian Railway links Moscow with the Pacific port of Vladivostok over about 9,289 km, crossing eight time zones. It was completed in 1916.
- 9. (A) Hawaii and California** — The patch lies in the North Pacific subtropical gyre between Hawaii and California, where circulating currents concentrate plastic waste. A global treaty on plastic pollution, mandated by the UN Environment Assembly in 2022, is still under negotiation.
- 10. (C) United States of America** — The United States has been the largest crude oil producer since 2018, ahead of Saudi Arabia and Russia. Saudi Arabia remains the largest exporter; OPEC, founded at Baghdad in 1960, has its headquarters at Vienna.
- 11. (D) The Panama Canal has locks while the Suez Canal has none** — The Suez Canal, opened in 1869 and 193 km long, is a sea-level canal joining the Mediterranean and the Red Sea and needs no locks. The Panama Canal, opened in 1914 and about 82 km long, lifts ships through the Gatun, Pedro Miguel and Miraflores locks.
- 12. (B) Hartsfield-Jackson Atlanta International** — Hartsfield-Jackson Atlanta has been the world's busiest airport by total passenger numbers for most of the last quarter century. Dubai handles the most international passengers and Hong Kong the most air cargo.
- 13. (A) Shanghai** — Shanghai has been the world's busiest container port since 2010, with Singapore second. Rotterdam is the largest port in Europe.
- 14. (C) Darien Gap** — The Darien Gap, on the Panama-Colombia border, is the only break in the Pan-American Highway, which otherwise runs from Alaska to southern Argentina.
- 15. (B) 1987** — The Montreal Protocol was adopted on 16 September 1987 and entered into force on 1 January 1989. It implements the Vienna Convention of 1985 and is the first treaty in UN history to secure universal ratification.
- 16. (D) hydrofluorocarbons** — The Kigali Amendment, adopted in Rwanda in 2016 and in force from 1 January 2019, phases down hydrofluorocarbons. HFCs do not deplete ozone but are powerful greenhouse gases. India ratified the amendment in 2021.
- 17. (C) 16 September** — 16 September marks the signing of the Montreal Protocol in 1987. World Environment Day is 5 June, World Wetlands Day 2 February and the World Day to Combat Desertification and Drought 17 June.
- 18. (A) the World Meteorological Organization and the United Nations Environment Programme** — The IPCC was set up by WMO and UNEP to assess the science of climate change. It shared the 2007 Nobel Peace Prize with Al Gore, and its Sixth Assessment Synthesis Report was released in 2023.
- 19. (B) COP21** — The Paris Agreement was adopted at COP21 in Paris on 12 December 2015 and entered into force on 4 November 2016. It replaced the Kyoto Protocol's imposed targets with Nationally Determined Contributions.
- 20. (D) well below 2 degrees Celsius above pre-industrial levels while pursuing efforts to limit it to 1.5 degrees Celsius** — Article 2 of the Paris Agreement sets the 'well below 2 degrees Celsius, pursuing 1.5 degrees Celsius' goal. Progress is reviewed through a global stocktake every five years, the first of which concluded at COP28 in 2023.
- 21. (A) 2070** — India announced the 2070 net-zero target at Glasgow in 2021 as part of the five-point 'Panchamrit' pledge. China's target year is 2060, and that of the European Union, the United Kingdom, the United States and Japan is 2050.
- 22. (C) COP27 in Sharm el-Sheikh** — COP27 at Sharm el-Sheikh in Egypt in 2022 agreed to create the Loss and Damage Fund. It was operationalised with the first pledges at COP28 in Dubai the following year.
- 23. (D) COP28 in Dubai** — COP28 at Dubai in 2023 concluded the first global stocktake, whose outcome for the first time called on countries to contribute to 'transitioning away from fossil fuels' and to treble renewable energy capacity by 2030.
- 24. (B) 300 billion US dollars a year by 2035** — COP29 (Baku, 2024) set a core goal of at least 300 billion US dollars a year by 2035 for developing countries, inside a wider 'Baku Finance Goal' aspiration of scaling all sources of climate finance to 1.3 trillion dollars a year by 2035.

- 25. (A) Belem** — COP30 met at Belem in the Brazilian Amazon; its main outcome decision is known as the 'Global Mutirao', which called for adaptation finance to be trebled by 2035. COP31 is to be held at Antalya in Türkiye with Australia holding the presidency.
- 26. (C) I, II and III** — All three are correct. New England around Boston pioneered American cotton textiles, Detroit is the automobile capital, and Pittsburgh grew where Appalachian coking coal met iron ore brought by the Great Lakes waterway.
- 27. (B) I only** — Sulphur dioxide and nitrogen oxides released by burning fossil fuels form sulphuric and nitric acid in the atmosphere and bring down rain with a pH below 5.6. Chlorofluorocarbons cause ozone depletion, not acid rain.
- 28. (D) I and II only** — The Montreal Protocol of 1987 implements the Vienna Convention of 1985 and has achieved universal ratification. Carbon dioxide is dealt with under the UNFCCC and its Kyoto Protocol and Paris Agreement, not under the Montreal Protocol.
- 29. (C) I, II and III** — All three are correct. The Kyoto Protocol was adopted at COP3 in 1997, entered into force on 16 February 2005 and covered the first commitment period 2008-2012; the Doha Amendment of 2012 extended it to 2020.
- 30. (A) I only** — The Paris Agreement rests on self-determined national pledges; the targets themselves are not legally binding and no absolute cut is imposed on developing countries. India's updated NDC promises a 45 per cent cut in the emission intensity of GDP by 2030 over 2005 levels.
- 31. (B) I and II only** — The UNCCD was adopted at Paris on 17 June 1994 and the day is therefore observed on 17 June, not 5 June. The Great Green Wall, launched by the African Union in 2007, aims to create an 8,000 km belt of restored land from Senegal to Djibouti.
- 32. (D) I, II and III** — All three are correct. A hotspot must in addition have lost at least 70 per cent of its original natural vegetation. Four hotspots extend into India - the Himalaya, Indo-Burma, the Western Ghats and Sri Lanka, and Sundaland (the Nicobar Islands).
- 33. (A) Both I and II** — Tropical forests, chiefly in the Amazon basin, the Congo basin and South-East Asia, lose far more area each year than temperate or boreal forests, and agricultural expansion - cattle ranching, soybean and oil palm - is the main cause.
- 34. (C) Donbas - Poland** — The Donbas or Donetsk basin, a great coal and steel region, lies in eastern Ukraine and not in Poland. Poland's corresponding coal-and-steel region is Silesia, centred on Katowice.
- 35. (B) Carbon dioxide** — Carbon dioxide is a greenhouse gas but does not destroy stratospheric ozone. Chlorofluorocarbons, halons, carbon tetrachloride, methyl chloroform and methyl bromide are the controlled ozone-depleting substances under the Montreal Protocol.
- 36. (D) Chlorofluorocarbons** — The Kyoto basket covers carbon dioxide, methane, nitrous oxide, hydrofluorocarbons, perfluorocarbons and sulphur hexafluoride, with nitrogen trifluoride added by the Doha Amendment. Chlorofluorocarbons are controlled instead under the Montreal Protocol.
- 37. (A) Canadian Pacific Railway - Australia** — The Canadian Pacific Railway runs across Canada from Montreal to Vancouver. Australia's great east-west line is the Trans-Australian Railway, on which the Indian Pacific service runs from Perth to Sydney.
- 38. (C) Nitrogen is a major greenhouse gas** — Nitrogen and oxygen, though they make up about 99 per cent of the atmosphere, are not greenhouse gases because their molecules do not absorb infra-red radiation. Without the natural greenhouse effect the earth's mean surface temperature would be about minus 18 degrees Celsius.
- 39. (B) A-3, B-4, C-2, D-1** — Lorraine is in France, the Midlands or Black Country around Birmingham in the United Kingdom, the Ruhr in Germany and the Donbas in Ukraine.
- 40. (D) A-4, B-3, C-1, D-2** — Ramsar 1971 (Ramsar, Iran), CITES 1973 (Washington), the Convention on Biological Diversity 1992 (Rio Earth Summit) and the Minamata Convention on Mercury 2013 (Kumamoto, Japan). Ramsar and CITES both entered into force in 1975.
- 41. (C) A-3, B-1, C-4, D-2** — COP26 met at Glasgow in 2021, COP27 at Sharm el-Sheikh in 2022, COP28 at Dubai in 2023 and COP29 at Baku in 2024. COP30 was held at Belem in Brazil in 2025.
- 42. (A) A-2, B-1, C-3, D-4** — Chile leads world copper output, Kazakhstan supplies about two-fifths of the world's uranium, China is the largest coal producer and Indonesia now dominates world nickel supply. India is the second largest producer of coal.

- 43. (B) A-2, B-3, C-1, D-4** — Magnitogorsk is the great Ural steel city of Russia, Anshan in Manchuria is China's oldest major steel centre, Kitakyushu (Yawata) is Japan's classic steel town and Katowice lies in the Upper Silesian industrial region of Poland.
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- 44. (D) A-2, B-3, C-4, D-1** — Basel (1989) governs the trans-boundary movement of hazardous wastes, Rotterdam (1998) the prior informed consent procedure for hazardous chemicals and pesticides, Stockholm (2001) persistent organic pollutants and Minamata (2013) mercury.
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- 45. (A) II, I, IV, III** — The standard ordering is carbon dioxide (about 60 per cent), methane (about 20 per cent), chlorofluorocarbons (about 14 per cent) and nitrous oxide (about 6 per cent). Ranked by global warming potential per molecule, however, the order is very nearly reversed.
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